Development of the

Phase 1 First-In-Human Study Protocol

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Agenda

- Guideline: MFDS (ICH) – Good Clinical Practice (GCP)
- ENHERTU® Case Study

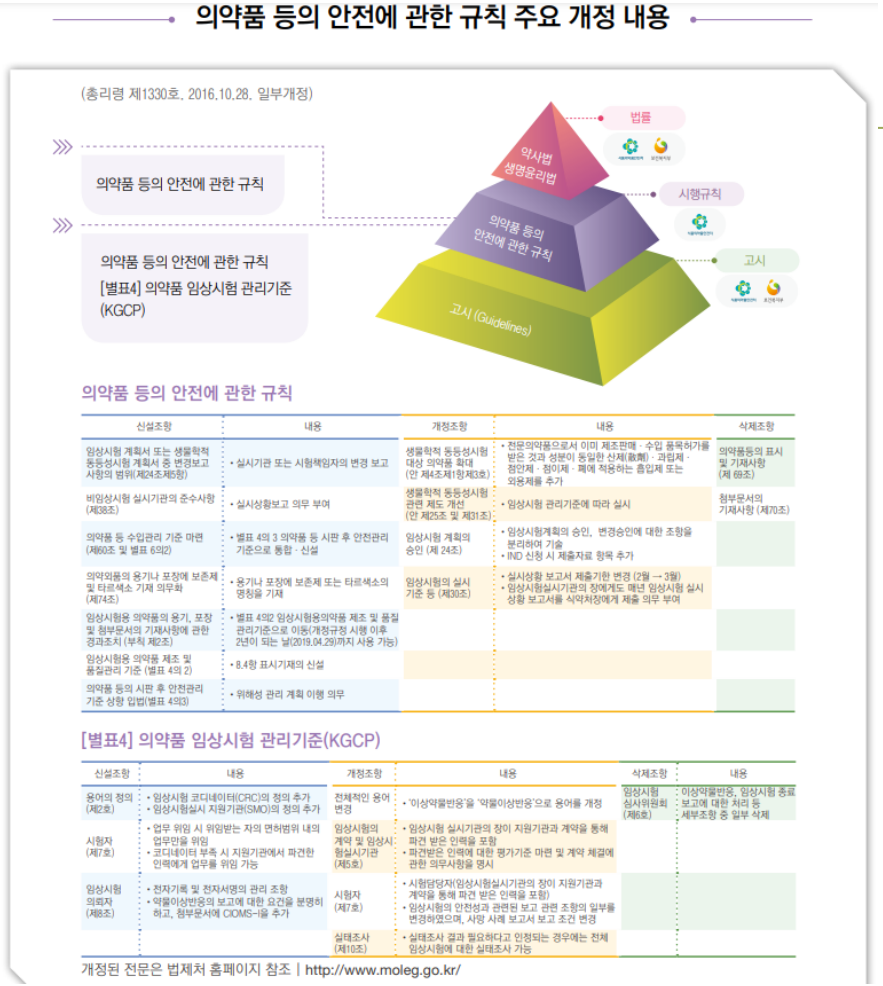


Regulatory Science

- 법률: 약사법

- 시행규칙:
  의약품 등의 안전에 관한 규칙
  (시행 2025-04-05)
  - 별표 4: **의약품 임상시험 관리 기준**
    (개정 2022-12-07)

- 고시:
  의약품 임상시험 계획 승인에 관한 규정
  (시행 2024-01-16)



의약품 등의 안전에 관한 규칙 주요 개정 내용

(총리령 제1330호, 2016.10.28. 일부개정)

의약품 등의 안전에 관한 규칙

의약품 등의 안전에 관한 규칙
[별표4] 의약품 임상시험 관리기준 (KGCP)

의약품 등의 안전에 관한 규칙

신설조항	내용	개정조항	내용	삭제조항
임상시험 계획서 또는 생물학적 동등성시험 계획서 중 변경보고 사항의 범위(제24조제5항)	• 실시기관 또는 시험책임자의 변경 보고	생물학적 동등성시험 대상 의약품 확대 (안 제4조제1항제3호)	• 전문의약품으로서 이미 제조판매 · 수입 품목허가를 받은 것과 성분이 동일한 산제(散劑) · 과립제 · 정인제 · 정이제 · 폐에 적응하는 흡입제 또는 외용제를 추가	의약품등의 표시 및 기재사항 (제 69조)
비임상시험 실시기관의 준수사항 (제38조)	• 실시상황보고 의무 부여	생물학적 동등성시험 관련 제도 개선 (안 제25조 및 제31조)	• 임상시험 관리기준에 따라 실시	첨부문서의 기재사항 (제70조)
의약품 등 수입관리 기준 마련 (제60조 및 별표 6의2)	• 별표 4의 3 의약품 등 시판 후 안전관리 기준으로 통합 · 신설	임상시험 계획의 승인 (제 24조)	• 임상시험계획의 승인, 변경승인에 대한 조항을 분리하여 기술 • IND 신청 시 제출자료 항목 추가	
의약외품의 용기나 포장에 보존제 및 타르색소 기재 의무화 (제74조)	• 용기나 포장에 보존제 또는 타르색소의 명칭을 기재	임상시험의 실시 기준 등 (제30조)	• 실시상황 보고서 제출기한 변경 (2월 → 3월) • 임상시험실시기관의 장에게도 매년 임상시험 실시 상황 보고서를 식약처장에게 제출 의무 부여	
임상시험용 의약품의 용기, 포장 및 첨부문서의 기재사항에 관한 경과조치 (부칙 제2조)	• 별표 4의2 임상시험용의약품 제조 및 품질관리 기준으로 이동(개정규정 시행 이후 2년이 되는 날(2019.04.29)까지 사용 가능)			
임상시험용 의약품 제조 및 품질관리 기준 (별표 4의 2)	• 8.4항 표시기재의 신설			
의약품 등의 시판 후 안전관리 기준 상향 입법(별표 4의3)	• 위해성 관리 계획 이행 의무			

[별표4] 의약품 임상시험 관리기준(KGCP)

신설조항	내용	개정조항	내용	삭제조항	내용
용어의 정의 (제2호)	• 임상시험 코디네이터(CRC)의 정의 추가 • 임상시험실시 지원기관(SMO)의 정의 추가	전체적인 용어 변경	• '이상약물반응'을 '약물이상반응'으로 용어를 개정	임상시험 심사위원회 (제8호)	• 이상약물반응, 임상시험 종료 : 보고에 대한 처리 등 세부조항 중 일부 삭제
시험자 (제7호)	• 업무 위임 시 위임받는 자의 면허범위 내의 업무만을 위임 • 코디네이터 부족 시 지원기관에서 파견한 인력에게 업무를 위임 가능	임상시험의 계약 및 임상시험실시기관 (제5호)	• 임상시험 실시기관의 장이 지원기관과 계약을 통해 파견 받은 인력을 포함 • 파견받은 인력에 대한 평가기준 마련 및 계약 체결에 관한 의무사항을 명시		
임상시험 의뢰자 (제8조)	• 전자기록 및 전자서명의 관리 조항 • 약물이상반응의 보고에 대한 요건을 분명히 하고, 첨부문서에 CIOMS-I을 추가	시험자 (제7호)	• 시험담당자(임상시험실시기관의 장이 지원기관과 계약을 통해 파견 받은 인력을 포함) • 임상시험의 안전성과 관련된 보고 관련 조항의 일부를 변경하였으며, 사망 사례 보고서 보고 조건 변경		
		실태조사 (제10조)	• 실태조사 결과 필요하다고 인정되는 경우에는 전체 임상시험에 대한 실태조사 가능		

개정된 전문은 법제처 홈페이지 참조 | http://www.moleg.go.kr/

[Reference] https://www.kscd.kr/web/board/newsLetterView.do?boardIdx=35

정의: 의약품 임상시험 관리 기준

- **임상시험 (Clinical Trial/Study)**: 임상시험용의약품의 안전성과 유효성을 증명할 목적으로, 해당 약물의 **약동 (藥動)・약력 (藥力)・약리 (藥理)**・임상 (臨床) 효과를 확인하고 이상반응을 조사하기 위하여 사람을 대상으로 실시하는 시험 또는 연구

- **임상시험 계획서 (Protocol)**: 해당 임상시험의 배경이나 근거를 제공하기 위하여 임상시험의 목적, 연구방법론, 통계적 고려사항, 관련 조직 등을 기술한 문서

- **임상시험 변경계획서 (Protocol Amendment, 변경계획서)**: 임상시험 계획서의 내용을 변경하거나 임상시험 계획서의 불분명한 부분을 명확하게 다시 기술한 문서

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

정의: 의약품 임상시험 관리 기준

- **임상시험용 의약품 (Investigational Product)**: 시험약 및 대조약

- 시험약: 임상시험용 의약품 중 대조약을 제외한 의약품

- **관련 규정 (Applicable Regulatory Requirement)**:
  임상시험의 실시에 관련된 약사법령 및 고시 등

- **임상시험의 준수 (Compliance, 준수)**: 계획서 및 관련 규정에 따라 임상시험을 실시하는 것

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

정의: 의약품 임상시험 계획 승인에 관한 규정

- **임상시험계획승인신청 (Investigational New Drug Application, IND)**:
  인체를 대상으로 한 안전성・유효성 자료 수집을 목적으로 해당 의약품을 사용하여 임상시험을 실시하고자 하는 자가 식품의약품안전처장 (식약처장)의 승인을 신청하는 과정

- **제1상 임상시험 (임상약리시험 등)**:
  내약성 평가, 약동학과 약력학 정의/서술, 약물대사와 상호작용 조사, 치료효과 추정을 목적으로 하는 임상 약리시험 등이며, 용량-내약성 임상시험, 단독과 반복 투여에 따른 약동학/약력학 임상시험, 약물 상호작용 임상시험 등이 해당

[Reference] https://www.konect.or.kr/kr/board/regulation_info/boardView.do?bbsIdx=3848&pageIndex=2&searchCondition=&searchKeyword=

의약품 등의 안전에 관한 규칙: 제24조 (임상시험계획의 승인 등)

- 임상시험계획의 승인을 받으려는 자는 별지 제23호서식의 임상시험계획 승인 신청서 (전자문서로 된 신청서를 포함한다)에 다음 각 호의 서류 (전자문서를 포함한다) 및 자료 (전자문서를 포함한다)를 첨부하여 식품의약품안전처장에게 제출해야 한다

① 개발계획

② **임상시험자 자료집**

③ 별표 1의 의약품 제조 및 품질관리기준 및 별표 4의2의 **임상시험용의약품 제조 및 품질관리기준에 맞게 제조되었음을 증명하는 서류 또는 자료** (평가받는 데 필요하다고 식품의약품안전처장이 정하여 고시하는 서류 또는 자료를 포함한다).
가. 생물학적제제등: 별표 3의 생물학적제제등 제조 및 품질관리기준

④ **임상시험용의약품 관련 제조 및 품질에 관한 자료**

⑤ **비임상시험성적에 관한 자료**

⑥ 시험약의 과거 임상적 사용경험에 관한 자료 (제출할 수 있는 경우만 해당한다)

⑦ 임상시험실시기관, 같은 항에 따른 임상시험검체분석기관, 시험자 및 수탁기관 등에 관한 자료

⑧ 임상시험 피해자 보상에 관한 규약

⑨ 시험대상자 동의서 서식

⑩ **임상시험 계획서**

임상시험 계획서에 포함되어야 할 사항

① 시험의 제목, 단계, 계획서 식별번호 및 제개정이력 등

② 시험계획서 요약

③ 서론 (배경, 이론적 근거, 유익성·위험성 평가 및 용량 설정 근거 등)

④ 시험의 목적

⑤ 시험모집단 (대상자수, 선정기준, 제외기준 및 중도탈락기준 등)

⑥ 시험 설계 내용 (시험기간, 시험군·대조군, 배정, 눈가림 및 흐름도 등)

⑦ 시험종료 및 조기중단 기준

⑧ 임상시험용의약품의 정보 및 관리 (표시 및 포장, 투여경로, 투여방법, 보관조건, 출납관리, 회수 및 폐기 등)

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

임상시험 계획서에 포함되어야 할 사항

⑨ 시험의 방법 및 투약계획 등 (투여 및 치료일정, 병용약물, 투여금지 약물 및 치료순응도 등)

⑩ 시험 절차 및 평가 (방문일정, 시험일정표, 유효성·안전성 평가변수와 평가 및 이상사례 보고 등)

⑪ 자료분석 및 통계학적 고려사항 (분석군, 통계분석방법, 판정기준, 분석시기 및 대상자수 설정근거 등)

⑫ 자료관리 (기록, 수집, 접근, 보호 및 보관 등)

⑬ 윤리적 고려사항 및 행정적 절차

(임상시험관리기준 및 동의절차 등 규정, 윤리준수, 대상자 안전보호 대책, 결과발표, 환자기록 비밀유지, 품질관리 및 신뢰성 보증 등)

⑭ 임상시험을 하려는 자 (임상시험 의뢰자)의 정보, 시험책임자 성명 및 직책

⑮ 그 밖에 임상시험을 안전하게 과학적으로 실시하기 위하여 필요한 사항

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

임상시험계획 변경: 변경 승인이 필요한 경우

① 시험 목적의 변경

② 신청인의 변경

③ **시험대상자의 안전** 또는 **시험결과의 신뢰성**에 **중대한 영향**을 미치는 변경

1) **시험모집단**의 변경으로서 대상자수의 유의한 변경, 대상자 선정기준의 변경, 대상자 제외기준의 변경 또는 중도탈락 기준의 완화

2) **시험설계**의 변경으로서 시험군 또는 대조군의 제외

3) **임상시험용의약품** 정보의 변경으로서 기준의 완화, 시험항목의 삭제 또는 제조원의 변경

4) **안전성·유효성 평가기준의 변경**으로서 평가변수의 변경

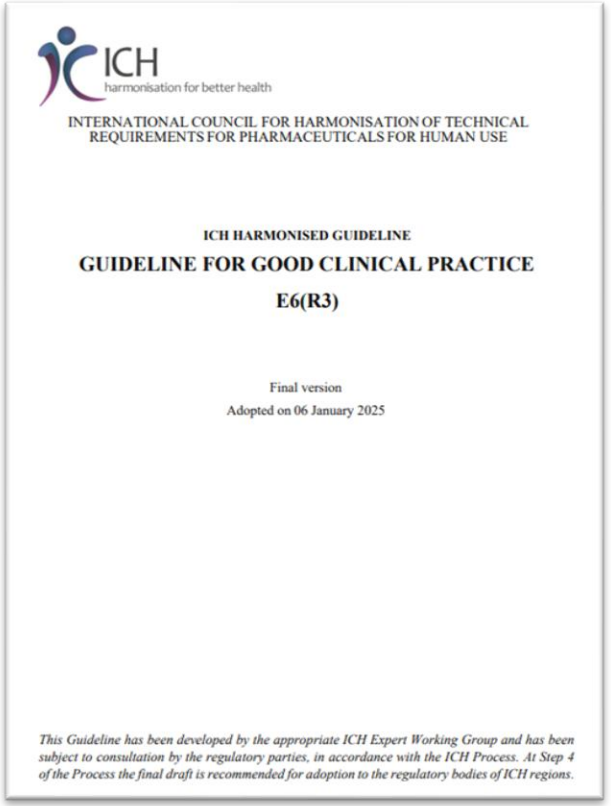
[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

Good Clinical Practice

- **ICH** (International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use)

- **GCP** (Good Clinical Practice)

- **E6 (R3)**: Final version, 2025-01-06



ICH
harmonisation for better health

INTERNATIONAL COUNCIL FOR HARMONISATION OF TECHNICAL REQUIREMENTS FOR PHARMACEUTICALS FOR HUMAN USE

ICH HARMONISED GUIDELINE

GUIDELINE FOR GOOD CLINICAL PRACTICE

E6(R3)

Final version

Adopted on 06 January 2025

This Guideline has been developed by the appropriate ICH Expert Working Group and has been subject to consultation by the regulatory parties, in accordance with the ICH Process. At Step 4 of the Process the final draft is recommended for adoption to the regulatory bodies of ICH regions.

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

GCP: ICH E6 (R3) Guideline

Appendix B. CLINICAL TRIAL PROTOCOL AND PROTOCOL AMENDMENT (S)

- Clinical trials should be described in a **clear**, **concise** and **operationally feasible** protocol
- The protocol should be designed in such a way as to **minimise unnecessary complexity** and to **mitigate or eliminate important risks to the rights, safety, and well-being of trial participants** **and the reliability of data**
- Protocol development processes should incorporate input from relevant interested parties, where appropriate
- Building adaptability into the protocol, for example, by including acceptable ranges for specific protocol provisions, **can reduce the number of deviations** or in some instances the requirement for a protocol amendment
- Such adaptability **should not adversely affect participant safety** or **the scientific validity of the trial**
- For additional information, refer to ICH E8 (R1) General Considerations for Clinical Studies, ICH E9 Statistical Principles for Clinical Trials and ICH E9 (R1) Addendum on Estimands and Sensitivity Analysis in Clinical Trials to the Guideline on Statistical Principles for Clinical Trials. The contents of a trial protocol should generally include the following topics, which may vary depending on the **trial design**. Investigator site-specific information may be provided on separate protocol page (s) or addressed in a separate agreement, and some of the information listed below may be contained in other protocol referenced documents, such as an Investigator's Brochure

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

1. General Information

① Protocol **title**, unique protocol identifying number and date. Any amendment (s) should also bear the amendment number (s) and date (s)

② Name and address of the **sponsor**

③ Name and title of the **person** (s) authorised to **sign the protocol** and the protocol amendment (s) for the sponsor

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

2. Background Information

① Name and description of the **investigational product**(s)

② A summary of **findings from nonclinical studies** that potentially have **clinical significance** and from clinical trials that are relevant to the trial

③ Summary of the **known and potential risks and benefits**, if any, to human participants

④ Description of and **justification for the route of administration, dosage, dosage regimen and treatment period(s)**

⑤ A statement that the trial will be conducted in compliance with the protocol, Good Clinical Practice (GCP) and the applicable regulatory requirement(s)

⑥ Description of the **population** to be studied

⑦ References to literature and data that are relevant to the trial and that provide background for the trial

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

3. Trial Objectives and Purpose

- A clear description of the scientific objectives and the purpose of the trial
- Information on estimands, when defined (see ICH E9 (R1))

4. Trial Design

- The scientific integrity of the trial and the reliability of the results from the trial substantially depend on the trial design
- A description of the trial design should include:

① A specific statement of the primary endpoints and the secondary endpoints, if any, to be measured during the trial

② A description of the type and design of trial to be conducted (e.g., double-blind, placebo-controlled, parallel design, adaptive design, platform/umbrella/basket, trials with decentralised elements) and a schematic diagram of trial design, procedures and stages

③ A description of the measures taken to minimise/avoid bias, including: (a) Randomisation (b) **Blinding**

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

4. Trial Design

④ A description of the **investigational product(s)** and the **dosage and dosage regimen** of the investigational product(s), including a description of the **dosage form, packaging** and **labelling**

⑤ **Preparation (e.g., reconstitution)** and **administration** instructions where applicable, unless described elsewhere

⑥ A description of the schedule of events (e.g., trial visits, interventions and assessments)

⑦ The expected duration of the participant's involvement in the trial and a description of the sequence and duration of all trial periods, including follow-up, if any

⑧ A description of the "**stopping rules**" or "**discontinuation criteria**" and "**dose adjustment**" or "**dose interruption**" for individual participants, for parts of the trial or for the entire trial

⑨ Accountability procedures for the **investigational product**(s), including the placebo(s) and other **comparator(s)**, if any

⑩ Maintenance of treatment **randomisation codes** and procedures for breaking codes

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

5. Selection of Participants

① Participant **inclusion** criteria

② Participant **exclusion** criteria

③ Mechanism for pre-screening, where appropriate, and screening of participants

6. Discontinuation of Trial Intervention and Participant Withdrawal from Trial

- The investigator may choose to discontinue the participant from the trial. Conversely, the participant may decide to withdraw from the trial or stop treatment with the investigational product
- The protocol should specify: (a) When and how to discontinue participants from the trial/investigational product treatment; (b) The type and timing of the data to be collected for withdrawn/discontinued participants, including the process by which the data are handled, in accordance with applicable regulatory requirements; (c) Whether and how participants are to be replaced; (d) The follow-up for participants who have discontinued the use of the investigational product

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

7. Treatment and Interventions for Participants

① The treatment(s) to be administered, including the name(s) of all the product(s), the dose(s), the dosing schedule(s), the criteria for dose adjustment(s), the route/mode(s) of administration and the treatment period(s), including the follow-up period(s) for participants for each investigational product treatment/trial treatment group/arm of the trial

② Medication(s)/treatment(s) permitted (including concomitant and rescue medication) and not permitted before and/or during the trial

③ Strategies to monitor the participant's adherence to treatment

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

8. Assessment of Efficacy

① Specification of the **efficacy parameters**, where applicable

② **Methods and timing for assessing, recording and analysing efficacy parameters**. Where any trial-related committees (e.g., independent data monitoring committee (IDMC)/adjudication committees) are utilised for the purpose of assessing efficacy data, the committees' procedures, timing and activities should be described in the protocol or a separate document

9. Assessment of Safety

① Specification of **safety parameters**

② **The methods, extent and timing for recording and assessing safety parameters**. Where any trial-related committees (e.g., IDMC) are utilised for the purpose of assessing safety data, procedures, timing and activities should be described in the protocol or a separate document

③ Procedures for obtaining reports of and for recording and reporting adverse events

④ **The type and duration of the follow-up of participants after adverse events and other events such as pregnancies**

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

10. Statistical Considerations

① A description of the statistical methods to be employed, including timing and purpose of any planned interim analysis(ses) and the statistical criteria for the stopping of the trial

② **The number of participants planned to be enrolled and the reason for the choice of sample size, including reflections on or calculations of the power of the trial and clinical justification**

③ The level of significance to be used or the threshold for success on the posterior probability in a Bayesian design

④ **The selection of participants** to be included in the planned analyses, a description of the statistical methods to be employed and procedures for handling intercurrent events and accounting for missing, unused and spurious data. These should be aligned with the target estimands, when defined (see ICH E9 (R1))

⑤ Statement that any deviation(s) from the statistical analysis plan will be described and justified in the clinical trial report

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

11. Direct Access to Source Records

- The sponsor should ensure that it is specified in the protocol or other documented agreement that the investigator(s)/institution (s)/service provider(s) will permit trial-related **monitoring, audits, regulatory inspection(s)** and, in accordance with applicable regulatory requirements, review by the institutional review board/independent ethics committee (IRB/IEC), providing direct access to source records

12. Quality Control and Quality Assurance

① Description of identified critical to quality factors, associated risks and risk mitigation strategies in the trial unless documented elsewhere

② Summary of the **monitoring** approaches that are part of the quality control process for the clinical trial

③ Description of the process for the handling of **noncompliance** with the protocol or GCP

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

Protocol

13. Ethics: Description of ethical considerations relating to the trial

14. Data Handling and Record Keeping

① Specification of data to be collected and the method of its collection. Where necessary, additional details should be contained in a clinical trial-related document

② The identification of data to be recorded directly into the data acquisition tools (i.e., no prior written or electronic record of data) and considered to be the source record

③ A statement that records should be retained in accordance with applicable regulatory requirements

15. Financing and Insurance: Financing and insurance, if not addressed in a separate agreement

16. Publication Policy: Publication policy, if not addressed in a separate agreement

[Reference] https://database.ich.org/sites/default/files/ICH_E6%28R3%29_Step4_FinalGuideline_2025_0106.pdf

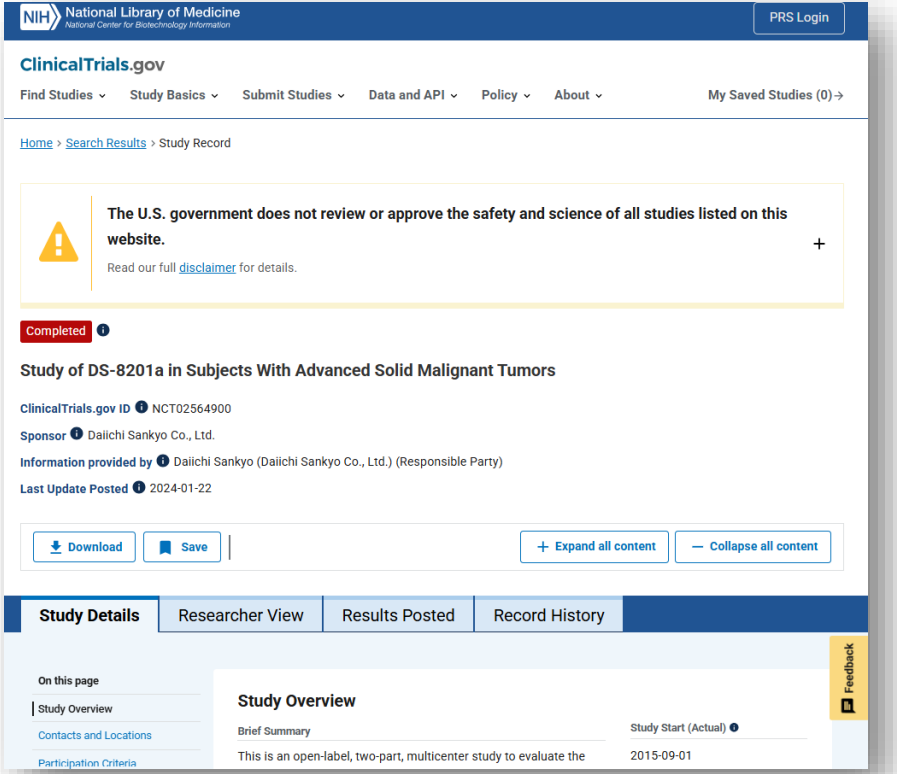
ENHERTU®: Clinicaltrials.gov (NCT02564900)

- **First-in-human Study: Study of DS-8201a in Subjects With Advanced Solid Malignant Tumors**
- **Sponsor:** Daiichi Sankyo Co., Ltd.

- **Brief Summary**

  This is an open-label, two-part, multicenter study to evaluate the safety and tolerability of DS-8201a in participants with advanced solid malignant tumors
- **Official Title**

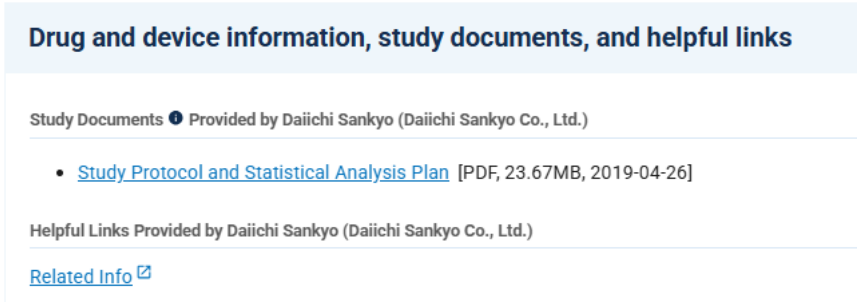
  Phase 1, Two-Part, Multicenter, Non-randomized, Open-label, Multiple Dose First-In-Human Study of DS-8201A, in Subjects With Advanced Solid Malignant Tumors



[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: Clinicaltrials.gov (NCT02564900)

- Drug and device information, study documents, and helpful links
  - Study Protocol & Statistical Analysis Plan





CLINICAL STUDY PROTOCOL

PHASE 1, TWO-PART, MULTICENTER, NON-RANDOMIZED, OPEN-LABEL, MULTIPLE DOSE FIRST-IN-HUMAN STUDY OF DS-8201A, IN SUBJECTS WITH ADVANCED SOLID MALIGNANT TUMORS

DS8201-A-J101

IND #: 127553

VERSION 12.0, 26 APR 2019

Version 1.0 (03 Jun 2015), Version 2.0 (26 Jun 2015), Version 3.0 (14 Jan 2016), Version 4.0 (29 Feb 2016), Version 5.0 (22 Jul 2016), Version 6.0 (09 Sep 2016), Version 7.0 (16 Nov 2016), Version 8.0 (07 Mar 2017), Version 9.0 (28 Apr 2017, Japan-specific), Version 10.0 (14 Sep 2017), Version 11.0 (25 Jan 2018)

DAIICHI SANKYO

CONFIDENTIALITY STATEMENT

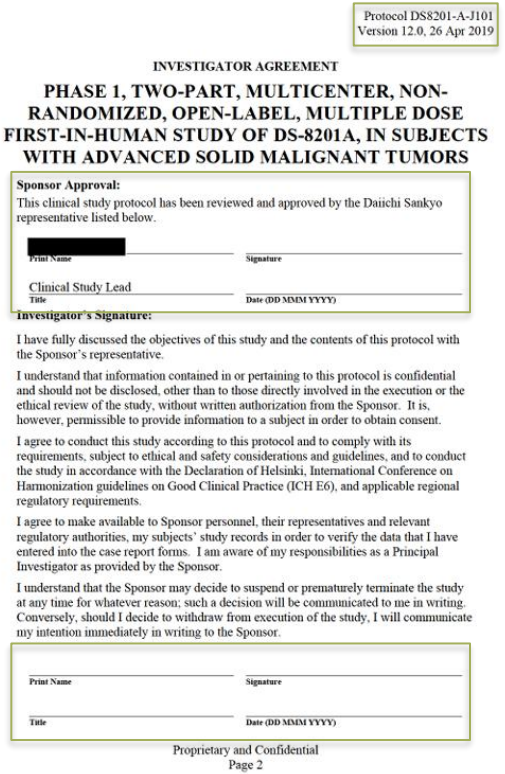
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Proprietary and Confidential
Page 1

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

- Protocol version
  - V1.0: 2015-06-03
  - V12.0: 2019-04-26
- Sponsor Approval
- Investigator’s Signature



Protocol DS8201-A-J101
Version 12.0, 26 Apr 2019

INVESTIGATOR AGREEMENT

PHASE 1, TWO-PART, MULTICENTER, NON-RANDOMIZED, OPEN-LABEL, MULTIPLE DOSE FIRST-IN-HUMAN STUDY OF DS-8201A, IN SUBJECTS WITH ADVANCED SOLID MALIGNANT TUMORS

Sponsor Approval:

This clinical study protocol has been reviewed and approved by the Daiichi Sankyo representative listed below.

[redacted] | Signature
Print Name

Clinical Study Lead | Date (DD MMM YYYY)
Title

Investigator's Signature:

I have fully discussed the objectives of this study and the contents of this protocol with the Sponsor's representative.

I understand that information contained in or pertaining to this protocol is confidential and should not be disclosed, other than to those directly involved in the execution or the ethical review of the study, without written authorization from the Sponsor. It is, however, permissible to provide information to a subject in order to obtain consent.

I agree to conduct this study according to this protocol and to comply with its requirements, subject to ethical and safety considerations and guidelines, and to conduct the study in accordance with the Declaration of Helsinki, International Conference on Harmonization guidelines on Good Clinical Practice (ICH E6), and applicable regional regulatory requirements.

I agree to make available to Sponsor personnel, their representatives and relevant regulatory authorities, my subjects' study records in order to verify the data that I have entered into the case report forms. I am aware of my responsibilities as a Principal Investigator as provided by the Sponsor.

I understand that the Sponsor may decide to suspend or prematurely terminate the study at any time for whatever reason; such a decision will be communicated to me in writing. Conversely, should I decide to withdraw from execution of the study, I will communicate my intention immediately in writing to the Sponsor.

Print Name | Signature

Title | Date (DD MMM YYYY)

Proprietary and Confidential
Page 2

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

- Protocol Synopsis
  - IND number
  - Protocol number
  - Investigational Product
  - Study title
  - Study phase
  - Indication
  - Objectives
  - Design

PROTOCOL SYNOPSIS

EudraCT/IND Number:	**Eudra CT: Not obtained/ IND Number: 127553**
Protocol Number:	DS8201-A-J101
Investigational Product:	DS-8201a
Active Ingredient(s)/INN:	Investigational new drug (INN) is not determined.
Study Title:	Phase 1, Two-part, Multicenter, Non-randomized, Open-label, Multiple Dose First-in-human Study of DS-8201a, in Subjects with Advanced Solid Malignant Tumors
Study Phase:	Phase 1
Indication Under Investigation:	DS-8201a will be evaluated in subjects with advanced solid tumors.
Study Objectives:	**Part 1 (Dose Escalation)** Primary Objectives: 1. To assess the safety and tolerability of DS-8201a. 2. To determine the maximum tolerated dose (MTD) or the recommended Phase 2 dose (RP2D) of DS-8201a. Secondary Objectives: 1. To assess the pharmacokinetic (PK) profile of DS-8201a, total anti-human epidermal growth factor receptor 2 (HER2) antibody, and MAAA-1181. 2. To evaluate the efficacy of DS-8201a. 3. To assess the incidence of anti-drug antibody (ADA) against DS-8201a. Exploratory Objectives: 1. To assess the biomarkers related to DS-8201a. **Part 2 (Dose Expansion)** Primary Objectives: 1. To assess the safety and tolerability of DS-8201a at the MTD/the RP2D. 2. To evaluate the efficacy of DS-8201a at the MTD/the RP2D. Secondary Objectives: 1. To assess the PK profile of DS-8201a, total anti-HER2 antibody, and MAAA-1181. 2. To assess the incidence of ADA against DS-8201a. Exploratory Objectives: 1. To assess the biomarkers related to DS-8201a.
Study Design:	This is a Phase 1, two-part, multicenter, non-randomized, open-label, multiple dose, first in human study of DS-8201a. This 2-part study will include both a Dose Escalation part, to identify the MTD or the RP2D of DS-8201a, and a Dose Expansion part, to confirm the safety, tolerability and efficacy of DS-8201a at the MTD/RP2D. Dose escalation of DS-

EudraCT/IND Number:	**Eudra CT: Not obtained/ IND Number: 127553**
	8201a to determine the MTD/RP2D will be guided by the modified continuous reassessment method (mCRM) using a Bayesian logistic regression model (BLRM) following escalation with overdose control (EWOC) principle. **Part 1 (Dose Escalation)** Dose escalation will begin in subjects with advanced breast cancer and gastric or gastroesophageal junction adenocarcinoma. At least 18 subjects will be enrolled in Part 1. More than 16% of subjects (eg, 1 per 6 subjects) are expected to be HER2 expression (immunohistochemistry [IHC] 2+ or 3+). The proposed human starting dose of DS-8201a is 0.8 mg/kg, based on 1/12 human equivalent dose (HED) of highest non-severely toxic dose (HNSTD) in monkey (30 mg/kg). During Part 1, an initial dose of DS-8201a will be infused intravenously into each subject for approximately 90 minutes on Day 1 of Cycle 1. A 21-day observation period (Cycle 1) will then occur, at the end of which all relevant safety data will be reviewed. Upon completion of Cycle 1, subjects may continue to receive DS-8201a once every 3 weeks (Q3W) (1 cycle) at the discretion of the Investigators, until unacceptable toxicity, progressive disease (PD), or withdrawal of consent. If there is no infusion related reaction after initial dose, the next dose of DS-8201a will be infused intravenously into each subject for approximately 30 minutes. The subject's weight at screening (baseline) will be used to calculate the initial dose. If the subject's weight changes by ± 10% of the baseline weight, the dose will be recalculated. **Dose level increment during dose escalation (Part 1) by mCRM with EWOC** – The dose level increment should be no less than 30% in order to have distinction among dose levels considering the inter-subject variability in exposure. – The dose level increment should be no more than 100% even if the model suggests a higher dose than 100% for the next cohort. The escalation by mCRM with EWOC principle will be based on a BLRM. The logistic regression model for the dose-response (dose limiting toxicity [DLT] rate) relationships will include 2 parameters, the intercept, and slope. After at least 3 subjects of each cohort complete DLT evaluation during Cycle 1, the posterior distributions of DLT rate are derived for all dose levels based on the BLRM using DLT outcome data from all assessed doses and a pre-specified prior distribution for the model parameters. The posterior probability of DLT rate in the following 4 intervals at each dose level: [0%, 16%] as

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

- Protocol Synopsis
  - Study duration
  - Sites & Location
  - Planned sample size

EudraCT/IND Number:	Eudra CT: Not obtained/ IND Number: 127553
	• AST or ALT > 5 × upper limit of normal (ULN), if accompanied by grade ≥ 2 blood bilirubin increased • In subjects without liver metastases, AST or ALT > 5 × ULN lasting >3 days • In subjects with liver metastases, AST or ALT > 5 × ULN lasting >3 days, if the baseline level was ≤ 3 × ULN • In subjects with liver metastases, AST or ALT > 8 × ULN lasting >3 days, if the baseline level was > 3 × ULN For non-hematological, non-hepatic major organ toxicities, a DLT is defined as follows: • Symptomatic Congestive heart failure (CHF) • Left ventricular ejection fraction (LVEF) decline to < 40% or > 20% drop from baseline • Other grade ≥ 3 non-hematological, non-hepatic major organ toxicities The following TEAEs are NOT considered DLTs: • Grade 3 fatigue lasting < 7 days • Grade 3 nausea, vomiting, diarrhea, or anorexia that has resolved to grade ≤ 2 within 3 days • Isolated laboratory findings not associated with signs or symptoms including grade 3/4 alkaline phosphatase (ALP) increased, hyperuricemia, serum amylase increased, and lipase increased, and grade 3 hyponatremia lasting < 72 hours developed from grade 1 at baseline • Grade 3 lymphocyte count decreased If any of the above toxicities is observed during the DLT evaluation period, whether or not the toxicity is regarded as DLT will be determined based on consultation between the Investigator and Sponsor. In addition, with regard to other toxicities that hinder the conduct of the scheduled study treatment or anemia with blood transfusion, whether or not they are regarded as DLT will be determined based on consultation between the Investigator and Sponsor.
Study Duration:	Part 1 is expected to last approximately one and a half years from the time the first subject is enrolled to the time the last subject is off the DLT evaluation period. Part 2 is expected to last approximately one and a half years from the time the first subject is enrolled to the time the last subject is off the study. The number of treatment cycles is not fixed in this study. Subjects who continue to derive clinical benefit from the study

EudraCT/IND Number:	Eudra CT: Not obtained/ IND Number: 127553
	treatment in the absence of withdrawal of consent, PD or unacceptable toxicity may continue the study treatment.
Study Sites and Location:	Study sites from Japan and the United States are planned for this study. Part 2e (Dose Expansion) will be conducted at Japanese sites only.
Planned Sample Size:	Part 1: At least 18 DLT-evaluable subjects are needed to reach an accurate estimate of the MTD. Part 2: Approximately 260 subjects (100 subjects for Part 2a, 40 subjects for Part 2b, 20 to 40 subjects for Part 2c, potentially additional 20 subjects for Part 2c, 60 subjects for Part 2d and 20 subjects for Part 2e) will be enrolled. Part 2a If target objective response rate (ORR) is more than 15% (null hypothesis: ORR ≤ 0.15, alternative hypothesis: ORR > 0.15), then the probability of less than 9 responders out of 100 subjects will be less than 5%. The probability that more than 21 responders out of 100 subjects (ORR > 21%) are observed will be less than 5% under the null hypothesis with ORR ≤ 0.15 but more than 90% under alternative hypothesis with ORR = 0.35. Part 2b If target ORR is more than 10% (null hypothesis: ORR ≤ 0.10, alternative hypothesis: ORR > 0.10), then the probability of no response out of 40 subjects will be less than 5%. The probability that more than 7 responders out of 40 subjects (ORR > 17.5%) are observed will be less than 5% under the null hypothesis with ORR ≤ 0.10 but more than 80% under alternative hypothesis with ORR = 0.25. Part 2c and Part 2e If target ORR is more than 15% (null hypothesis: ORR ≤ 0.15, alternative hypothesis: ORR > 0.15), then the probability of no response out of 20 subjects will be less than 5%. The probability that more than 4 responders out of 20 subjects (ORR > 20%) are observed will be less than 20% under the null hypothesis with ORR ≤ 0.15 but more than 75% under alternative hypothesis with ORR = 0.30. Part 2d If target ORR is more than 15% (null hypothesis: ORR ≤ 0.15, alternative hypothesis: ORR > 0.15), then the probability of less than 5 responders out of 60 subjects will be less than 5%. The probability that more than 14 responders out of 60 subjects (ORR > 23.3%) are observed will be less than 5%

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

- Protocol Synopsis
  - Eligibility criteria

    (Inclusion & Exclusion)

EudraCT/IND Number:	Eudra CT: Not obtained/ IND Number: 127553
	under the null hypothesis with ORR ≤ 0.15 but more than 80% under alternative hypothesis with ORR = 0.30. The probability values for the sample size are derived based on binomial distribution using SAS® version 9.2.
Subject Eligibility Criteria:	**Inclusion Criteria** Subjects must satisfy all of the following criteria to be included in the study: **Common Inclusion Criteria (Part 1 and Part 2):** 1. Age ≥ 20 years in Japan, ≥ 18 years in the United States. 2. Has an Eastern Cooperative Oncology Group performance status (ECOG PS) 0-1. 3. Has LVEF ≥ 50% by either echocardiography (ECHO) or multiple-gated acquisition (MUGA) within 28 days before registration. 4. Has adequate organ function within 7 days before registration, defined as:

Item	Laboratory value
Platelet count	≥ 100 000/mm³
Hemoglobin (Hb)	≥ 8.5 g/dL
Absolute neutrophil count (ANC)	≥ 1500/mm³
Creatinine	≤ 1.5 × ULN, or creatinine clearance ≥ 60 mL/min as calculated using the modification Cockcroft-Gault equation
AST/ALT	≤ 3 × ULN (if liver metastases are present, ≤ 5 × ULN)
Total bilirubin	≤ 1.5 × ULN or < 3 × ULN in the presence of documented Gilbert's Syndrome or liver metastases at baseline
Prothrombin time and activated partial thromboplastin time	≤ 1.5 × ULN

5. Has adequate treatment washout period before registration, defined as:

Treatment	Washout period
Major surgery	≥ 4 weeks
Radiation therapy	≥ 4 weeks (if palliative stereotactic radiation therapy without abdominal, ≥ 2 weeks)
Autologous transplantation	≥ 3 months
Hormonal therapy	≥ 3 weeks
Chemotherapy	≥ 3 weeks (≥ 2 weeks for 5-fluorouracil-based agents, folinate agents and/or weekly Paclitaxel. ≥ 6 weeks for nitrosoureas or mitomycin C. >1 week for TKIs approved for the treatment of NSCLC - baseline CT scan must be completed after discontinuation of TKI)
Immunotherapy	≥ 4 weeks
Cytochrome P450 (CYP) 3A4 strong inhibitor	≥ 3 elimination half-lives of the inhibitor
Organic anion transporting polypeptide inhibitor	≥ 3 elimination half-lives of the inhibitor

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6. Is able to provide written informed consent. Subject must be fully informed about their illness and the investigational nature of the study protocol (including foreseeable risks and possible toxicities) and must sign and date an Institutional Review Board (IRB) approved informed consent form (ICF) (including Health Insurance Portability and Accountability Act [HIPAA] authorization, if applicable) before performance of any study-specific procedures or examinations.
7. Is willing to provide pre-existing diagnostic or resected tumor samples.
8. Has a life expectancy of ≥ 3 months.

Additional Inclusion Criteria for Part 1 (Dose Escalation)

9. Has a pathologically documented advanced/unresectable or metastatic breast cancer or gastric or gastroesophageal junction adenocarcinoma that is refractory to or intolerable with standard treatment, or for which no standard treatment is available.

Additional Inclusion Criteria for Part 2a (Dose Expansion)

10. Has a pathologically documented advanced/unresectable or metastatic breast cancer with HER2 overexpression (IHC 3+ or in situ hybridization [ISH*] +) that is refractory to or intolerable with standard treatment, or for which no standard treatment is available.
    * ISH: fluorescence in situ hybridization (FISH) or dual color in situ hybridization (DISH)
11. Treated with T-DM1.
12. Has measurable disease based on Response

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

- Protocol Synopsis
  - Dosage Form, Dose & Route of Administration
  - Study Endpoints
    - ✓ Safety
    - ✓ PK
    - ✓ Efficacy
  - Statistical Analysis

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	[for US subjects]) during the study and for at least 7 months after the last dose of DS-8201a. 9. Has clinically active brain metastases, defined as untreated and symptomatic, or requiring therapy with steroids or anticonvulsants to control associated symptoms. Subjects with treated brain metastases that are no longer symptomatic and who require no treatment with steroids may be included in the study if they have recovered from the acute toxic effect of radiotherapy. 10. Has unresolved toxicities from previous anticancer therapy, defined as toxicities (other than alopecia) not yet resolved to NCI-CTCAE version 4.0, grade ≤ 1 or baseline. Subjects with chronic grade 2 toxicities may be eligible per the discretion of the Investigator. 11. Has a concomitant medical condition that would increase the risk of toxicity, in the opinion of the Investigator. 12. Has known hypersensitivity to either the drug substances or inactive ingredients in the drug product. 14. Has a history of severe hypersensitivity reactions to other monoclonal antibodies. **Additional Exclusion Criteria for Part 2 (Dose Expansion)** 13. Has multiple primary malignancies within 3 years, except adequately resected non-melanoma skin cancer, curatively treated in-situ disease, other solid tumors curatively treated, or contralateral breast cancer.
Dosage Form, Dose and Route of Administration:	Two types of DS-8201a drug product, DS-8201a for Injection [redacted] (FL-DP1) and DS-8201a for Injection [redacted] (FL-DP2), will be supplied in this study. FL-DP1 material will be used for Part 1, Part 2a, 2b, 2c and 2d. FL-DP2 material will be used for Part 2e. DS-8201a will be administered as an IV solution. Subjects with solid malignant tumors will treat with DS-8201a on Day 1 of Q3W. An alternative drug administration schedule for dose expansion may be considered after establishing the MTD and/or RP2D of DS-8201a using the above schedule, and upon review of the human safety, and PK parameters by the Investigators and Sponsor. RP2D Determined RP2D of DS-8201a is 6.4 mg/kg Q3W or 5.4 mg/kg Q3W. In Part 2a, approximately 50 subjects will receive 6.4 mg/kg of DS-8201a Q3W and another approximately 50 subjects will receive 5.4 mg/kg of DS-8201a Q3W. In Part 2b, approximately 20 subjects will receive 6.4

EudraCT/IND Number:	Eudra CT: Not obtained/ IND Number: 127553
	mg/kg of DS-8201a Q3W and another approximately 20 subjects will receive 5.4 mg/kg of DS-8201a Q3W. In Part 2c, first approximately 20 subjects will receive 6.4 mg/kg of DS-8201a Q3W but additional approximately 20 subjects may receive another dose level of DS-8201a Q3W. In Part 2d and Part 2e, all subjects will receive 6.4 mg/kg of DS-8201a Q3W.
Study Endpoints:	Safety endpoints: Safety endpoints will include DLTs, serious adverse events (SAEs), TEAEs, physical examination findings (including ECOG PS), vital sign measurements, standard clinical laboratory parameters, ECG parameters, ECHO/MUGA findings, and ophthalmologic findings. TEAEs will be graded according to the NCI-CTCAE version 4.0. Pharmacokinetic endpoints: Serum PK parameters (area under the concentration-versus-time curve, from time 0 to the last measurable concentration as calculated by the linear-up log-down trapezoidal method [AUClast], AUC during dosing interval [AUCtau], maximum (peak) observed concentration [Cmax], time of maximum observed concentration [Tmax] and trough serum concentrations [Ctrough]) of DS-8201a for each subject will be estimated using standard noncompartmental methods. For total anti-HER2 antibody and MAAA-1181, all parameters listed above will be estimated. Descriptive statistics will be provided for all serum concentration data and PK parameter values, with a breakdown by dose level/cohort within each Part and study day as appropriate. Efficacy endpoints: Tumor response will be evaluated using RECIST version 1.1. ORR (the sum of complete response [CR] rate and partial response [PR] rate), disease control rate (DCR, the sum of CR rate, PR rate, and stable disease [SD] rate), response duration, duration of SD, time to response (TTR), and progression free survival (PFS), overall survival (OS), percent change in target lesion, time on therapy of the most recent prior regimen the subject received and that of DS-8201a. The Efficacy Variable(s) will be also evaluated at 18 weeks after Day 1 of Cycle 1.
Statistical Analyses:	The data cutoff for the primary analysis will occur after all subjects have either discontinued the study or the last subject enrolled in Part 2 of the study has completed at least 6 months of study drug treatment. After the primary analysis, the main study will be closed and the data will be followed until completion. Descriptive statistics will be provided for selected demographic, safety, and PK data by dose level/cohort within

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Non-clinical Studies

Population, Route, Dosage, Dosage Regimen, Treatment Period

Dose-Limiting Toxicities

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PK, PD Biomarker

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Lab information

HER2 Test : IHC, FISH, Cell-free DNA

Immune Profiling

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Sampling Timepoints

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1. INTRODUCTION AND BACKGROUND INFORMATION

1.1. Data Summary

1.1.1. Investigational Product(s)

1.1.1.1. Name

DS-8201a

1.1.1.2. Description

DS-8201a is an antibody-drug conjugate (ADC) that targets HER2. DS-8201a consists of an antibody component ("MAAL-9001" hereafter), which is a humanized immunoglobulin G1 (IgG1) monoclonal antibody produced in house with reference to the amino acid sequence of trastuzumab, that is bound via a maleimide tetrapeptide linker to a drug component, which is an exatecan derivative that is a topoisomerase I inhibitor ("MAAA-1181" hereafter).

Two types of DS-8201a drug product, DS-8201a for Injection [redacted] (FL-DP1 hereafter) and DS-8201a for Injection [redacted] (FL-DP2 hereafter), will be supplied in this study. FL-DP2 drug product is added to be evaluated in subjects enrolled in the newly added cohort of Part 2e.

1.1.1.3. Intended Use Under Investigation

DS-8201a will be evaluated in subjects with advanced solid tumors

1.1.1.4. Nonclinical Studies

1.1.1.4.1. Pharmacology

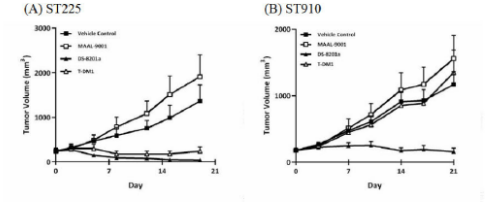
The fact that DS-8201a binds specifically to the HER2 extracellular domain (ECD) and does not bind to other HER family proteins has been confirmed by enzyme-linked immunosorbent assay (ELISA) using recombinant proteins. In addition, since the binding affinity of DS-8201a to HER2 was comparable to that of MAAL-9001, the antibody component of DS-8201a – a humanized IgG1 monoclonal antibody produced in-house with reference to the same amino acid sequence of trastuzumab, it is not considered that the conjugation of MAAA-1181 with MAAL-9001 has any effect on the binding affinity of MAAL-9001.

The results of in vitro cell growth inhibition studies conducted using several cancer cell lines have shown that DS-8201a has a more potent growth inhibition against HER2-positive cells than MAAL-9001, suggesting that the conjugation of MAAA-1181 enhances the growth inhibition of DS-8201a. Moreover, no growth inhibition was confirmed in HER2-negative cells, thus confirming the HER2 specificity of DS-8201a.

Similarly, when the in vivo antitumor activity of DS-8201a in a tumor-bearing mouse model of a HER2-positive gastric cancer cell line was studied, it was confirmed that DS-8201a exhibited potent, dose-dependent antitumor activity with tumor regression, and that this activity was even stronger than that of MAAL-9001.

In studies conducted in a tumor-bearing mouse model utilizing grafted tumor tissue obtained from breast cancer patients, DS-8201a exhibited antitumor activity both in an HER2 high-expressing model with an IHC score of 2+ and also in an HER2 low-expressing model with an IHC score of +1, in which T-DM1 was insensitive. Therefore, in the clinical setting, DS-8201a is expected to be effective against HER2 low-expressing tumors for which T-DM1 does not have an indication. (Figure 1.1)

Figure 1.1: Anti-tumor Effect of DS-8201a, MAAL-9001, and T-DM1 against ST225 and ST910 Patient-Derived Breast Cancer Xenograft Model



Data represent the mean + standard error of the mean (n = 5).
Mice were subcutaneously implanted with ST225 and ST910 patient-derived xenografts.
DS-8201a, MAAL-9001, or T-DM1 at the dose of 10 mg/kg were administered on Day 0.
The tumor volume of each mouse was calculated according to the following equation:
Tumor volume (mm^3) = 0.52 × length × $width^2$
HER2 expressing levels; ST225 model was IHC 2+ and FISH positive, and ST910 model was IHC 1+ and FISH negative.

In studies on the mechanism of action of DS-8201a, DS-8201a [redacted] has also been confirmed to cause deoxyribonucleic acid damage and induce apoptosis, effects that are assumed to be the result of MAAA-1181, which has topoisomerase I inhibitory activity.

Therefore, DS-8201a is considered to exhibit HER2-specific cell growth inhibition and antitumor activity via a novel mechanism of action that combined the pharmacological activities of MAAL-9001, the antibody component, with those of MAAA-1181, the drug component.

1.1.1.4.2. Safety Pharmacology

In telemetered male cynomolgus monkeys treated with single intravenous doses of DS-8201a, no effects on the cardiovascular system, the respiratory system, or the central nervous system were observed at dose levels up to 78.8 mg/kg. In addition, in human ether-a-go-go-related gene (hERG) studies of MAAA-1181, the drug component, MAAA-1181 did not inhibit the hERG channel current at concentrations of up to 10 µmol/L (approximately 5000 ng/mL).

Pharmacology (Patient-derived)

Safety Pharmacology

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1.1.1.4.3. Pharmacokinetics and Drug Metabolism

The plasma DS-8201a concentrations decreased exponentially following a single intravenous administration of DS-8201a at 0.1 mg/kg to 3.0 mg/kg to cynomolgus monkeys. The Vss was close to the plasma volume. The CL decreased as the dose increased, and the pharmacokinetics were found to be non-linear. Both DS-8201a and the total antibody, bound and unbound antibody combined, exhibited similar plasma concentration-time profiles at all dose levels, as well as similar AUC. All individual plasma concentrations of MAAA-1181, the released drug from DS-8201a, were below the lower limit of quantification (0.100 ng/mL) at 0.1 and 0.3 mg/kg. A low-plasma level of MAAA-1181 was detected at limited time points at 1.0 and 3.0 mg/kg. No anti-DS-8201a antibody was detected in any animals.

The mean plasma protein binding ratios of MAAA-1181 at 10 ng/mL to 100 ng/mL were from 90.3% to 92.5% in mice, 94.2% to 96.7% in rats, 86.5% to 89.1% in monkeys, and 96.8% to 98.0% in humans.

The release rates of MAAA-1181 from DS-8201a increased gradually throughout the incubation period in mouse, rat, and monkey plasma with the release rates from 1.2% to 3.9% on Day 21. On the other hand, the release rate reached a plateau on Day 14 in human plasma with the release rates from 2.2% to 2.4%. These results indicate that most DS-8201a is stable in plasma.

No major differences were found among the metabolite profiles of DS-8201a in rat, monkey, and human hepatocytes. MAAA-1181 was metabolized by CYP enzymes; CYP3A4 was the primary CYP enzyme in the metabolism, CYP3A5 and CYP2D6 were also involved in the metabolism.

The plasma concentration time profiles following repeated administration of DS-8201a Q3W for 3 cycles (Q3W × 3) to humans were simulated on the basis of the pharmacokinetics of DS-8201a in cynomolgus monkeys. These estimates were compared to the plasma concentrations of DS-8201a in the studies in tumor-bearing mice. As a result, the minimum effective dose and the pharmacologically active dose (PAD) were expected to be 0.8 and 4.8 mg/kg, respectively, with a dosage Q3W in humans.

1.1.1.4.4. Toxicology

Toxicology

In a study of intermittent intravenous dosing of DS-8201a in rats (Q3W dosing for 6 weeks), no deaths or moribund animals were found at dose levels up to 197 mg/kg, the maximum dose. The major observed findings included testicular and intestinal toxicity at dose levels of 20 mg/kg and greater, and lymphatic/hematopoietic, skin, and incisor tooth, and renal toxicity at dose levels of 60 mg/kg and greater. Except for the testicular and incisor tooth changes, these changes were all found to recover.

In an intermittent intravenous dosing study of DS-8201a in cynomolgus monkeys (Q3W, 6 weeks), one female was sacrificed moribund at 78.8 mg/kg, the highest dose level. The major toxicity findings in this moribund animal were observed in the intestine, hematopoietic system, skin, and kidney. The cause of the moribundity appeared to be the deteriorated condition of the animal which included decreased body weight and food consumption, as well as bone marrow toxicity and intestinal toxicity. The major findings of toxicity in the surviving animals were observed in the intestine at dose levels of 10 mg/kg and greater, and in the lung, testes, and skin at dose levels of 30 mg/kg and greater. In addition, hematopoietic system toxicity, renal toxicity,

and ECG abnormalities (shortened PR interval and QTc prolongation) were found at 78.8 mg/kg. Except for the pulmonary and skin toxicity (pigmentation), these findings tended to recover.

Thus, as described above, the severely toxic dose in 10% of the animals (STD_{10}) in a rat intermittent intravenous dosing study of DS-8201a was found to be greater than 197 mg/kg. In the monkey study, due to observed moribundity at 78.8 mg/kg and evidence of critical pulmonary toxicity (eg, interstitial inflammation and/or alveolar edema) in the surviving animals, it was concluded that the HNSTD is 30 mg/kg.

In an intermittent intravenous dose toxicity study of MAAA-1181 (qw dosing for 4 weeks), findings in the lymphatic/hematopoietic system, intestinal tract, and the cornea of the eye were observed at 3 mg/kg and greater in rats and there was no death or moribundity at up to 30 mg/kg. Findings similar to those in rats were observed in cynomolgus monkeys at dose levels of 1 mg/kg and greater. In addition, 1 female monkey died and 1 male monkey was sacrificed moribund at 12 mg/kg. Although effects on the heart (focal myocardial cell degeneration/necrosis) were found in the moribund male along with the above mentioned toxicities, there were no abnormal heart findings in the female that died even though both animals exhibited worsening clinical conditions associated with sustained decreases in food consumption, bone marrow toxicity, and intestinal toxicity. These changes were considered to be the cause of the death and moribundity. The common adverse findings with both DS-8201a and MAAA-1181 studies were intestinal and lymphatic/hematopoietic system toxicities. For DS-8201a treatment, pulmonary, testicular, skin and renal toxicities were observed while heart, liver, and corneal toxicities were found only when MAAA-1181 was administered.

In a human cross-reactivity study of DS-8201a with a panel of human tissues, DS-8201a-related cell membrane staining was found only in the placenta. In a cross-reactivity study of DS-8201a with selected cynomolgus monkey tissues (eg, brain, liver, kidney, lung, heart, intestines, lymphoid organs, testis, and skin), neither membranous nor cytoplasmic staining was noted in any tissues.

In an in vitro 3T3 NRU phototoxicity study, MAAA-1181 was found to be phototoxic to Balb/c 3T3 mouse fibroblasts. However, in an in vivo single dose phototoxicity study with MAAA-1181 in pigmented rats, no phototoxic reaction was noted at 3 mg/kg, the highest dose tested.

1.1.1.4.5. Human Starting Dose

Human Starting Dose

The initial human starting dose is determined by using the HNSTD in cynomolgus monkeys, which exhibit a higher level of toxicological sensitivity than rats and which exhibit a level of cross-reactivity similar to that of humans. For the HNSTD in cynomolgus monkeys, the HED is used. It was converted so that the dose per body surface area (mg/m^2) was the same in the experimental animals and in humans, in accordance with the extrapolation method that is widely used for drug products that exhibit cytotoxicity.

The HED of HNSTD was calculated to be 9.7 mg/kg by dividing the HNSTD in cynomolgus monkeys, 30 mg/kg, by the correction factor for cynomolgus monkeys of 3.1. In accordance with the Nonclinical Evaluation for Anticancer Pharmaceuticals (the ICH S9 guidelines), the estimated maximum starting dose for a Phase 1 clinical study would be 1.60 mg/kg as one-sixth of the HED of HNSTD. However, as a conservative approach, the human starting dose of 0.8

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

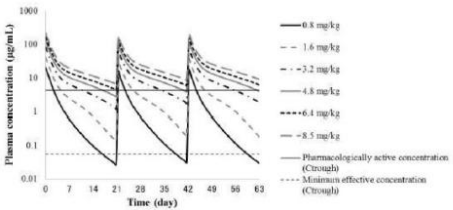
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Non-clinical PK/PD Modeling & Simulation

mg/kg which is one-twelfth of HED of HNSTD is selected for this study, because the results of the toxicity studies in cynomolgus monkeys suggested that the lungs might be affected.

Additionally, in the nonclinical pharmacology study, the plasma concentration-time profiles following repeated administration of DS-8201a Q3W for 3 cycles (Q3W × 3) to humans were simulated on the basis of the pharmacokinetics of DS-8201a in cynomolgus monkeys. The results are shown in Figure 1.2. Based on the Ctrough data during the treatment cycle, 0.8 and 4.8 mg/kg of the dose given Q3W would be required to achieve Ctrough close to the anticipated minimum effective concentration (0.0551 μg/mL) and pharmacologically active concentration (4.26 μg/mL). Assuming DS-8201a pharmacokinetics in humans are similar to those in cynomolgus monkeys, these results suggest that the minimum effective dose and the PAD were expected to be 0.8 and 4.8 mg/kg, respectively, with a dosage Q3W in humans.

Figure 1.2: Predicted Plasma Concentration Profiles of DS-8201a in Monkeys after Repeated Intravenous Administration Every 3 Weeks for 3 Cycles



Minimum effective concentration and pharmacologically active concentration were determined in a human gastric tumor mouse xenograft model.

1.1.1.5. Clinical Experience

This is the first-in human study of DS-8201a. No prior clinical experience of DS-8201a is available.

1.2. Study Rationale

DS-8201a is an ADC that targets HER2. A humanized IgG1 monoclonal antibody with reference to the amino acid sequence of trastuzumab is used as the antibody component and a derivative of exatecan, a topoisomerase I inhibitor, is used as the drug component. DS-8201a is expected to inhibit tumor growth on the basis of the following reasons: it exhibits [redacted] the MAAA-1181 that is released from DS-8201a after the internalization induces apoptosis by inhibiting topoisomerase I.

HER2 is a member of the HER superfamily, and initiates signal transduction via the PI3K/AKT and RAS/MAPK pathways.[1, 2] In human advanced solid tumors, expression of HER2 protein has been reported in various tumor tissues and a variety of cultured tumor cell lines including breast cancer,[3] gastric cancer,[4, 5] pancreatic cancer,[6] lung cancer,[7] colorectal cancer,[8] and ovarian cancer.[9] There are also many reports demonstrating an association between expression of HER2 protein and clinical poor prognosis. In normal human tissue, low expression of HER2 protein has been reported on cell membranes of epithelial cells in the gastro-intestinal, respiratory, reproductive, and urinary tract as well as in the skin, breast and placenta.[10]

As an antibody targeting HER2, Trastuzumab has been approved in the United States for the indication of HER2 overexpressing breast cancer and HER2-overexpressing metastatic gastric or gastroesophageal junction adenocarcinoma,[11] in Europe for HER2 positive metastatic breast cancer and HER2 positive metastatic adenocarcinoma of the stomach or gastroesophageal junction,[12] and in Japan for HER2-overexpressing breast cancer and HER2-overexpressing unresectable or advanced/recurrent gastric or gastroesophageal junction adenocarcinoma.[13] T-DM1 is an ADC targeting HER2 that has been approved in the United States for the indication of HER2-positive metastatic breast cancer,[14] in Europe for the indication of HER2-positive unresectable locally advanced or metastatic breast cancer,[15] and in Japan for HER2-positive unresectable or advanced breast cancer.[16]

In nonclinical studies, the fact that DS-8201a binds specifically to the HER2 ECD and does not bind to other HER family proteins has been confirmed. In vitro studies indicate that DS-8201a exhibits HER2 expression-dependent cell growth inhibition activity, moreover, no growth inhibition was confirmed in HER2-negative cells, thus confirming the HER2 specificity of DS-8201a. In vivo studies using a tumor-bearing mouse model suggest that administration of DS-8201a results in the regression of HER2-positive tumors. Emtansine, which has a tubulin polymerization inhibition activity, is used as the drug component in T-DM1. Since MAAA-1181 that inhibits topoisomerase I is used in DS-8201a, DS-8201a is expected to be effective even in tumors that do not respond or are resistant to T-DM1. DS-8201a exhibits anti-tumor effect with tumor regression in tumors that express low levels of HER2, against which T-DM1 has little effect, and is therefore also expected to be effective against tumors that express low levels of HER2, for which HER2 therapy including T-DM1 has not been approved. One reason for this may be that the drug-to-antibody ratio (DAR) for DS-8201a is approximately 8, compared to a DAR of average 3.5 for T-DM1,[14] meaning that DS-8201a can deliver more drugs into the cell than T-DM1.

In summary, DS-8201a has a different drug component from that of T-DM1 and is expected to show activity in the tubulin inhibitor insensitive tumors and in low HER2-expressing tumors. Thus, DS-8201a will be a benefit to a broader patient population than T-DM1. Therefore, DS-8201a is being developed as an ADC targeting HER2. This clinical trial is the first-in-human study designed to evaluate the safety, tolerability, PK, and efficacy of DS-8201a doses using FL-DP1 drug product in the dose escalation Part 1 of the trial; after the dose for expansion is selected from Part 1, the safety, tolerability, efficacy and PK, will be further evaluated in dedicated cohorts of subjects with solid tumors administered FL-DP1 drug product (Parts 2a, 2b, 2c, and 2d) and in subjects administered FL-DP2 drug product (Part 2e). The PK profile between FL-DP1 and FL-DP2 drug product will be assessed for bridging in clinical use.

Study Rationale

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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Risks & Benefits

Route Dose

1.3. Risks and Benefits for Study Subjects

Non-clinical studies have demonstrated the potent anti-tumor activity of DS-8201a in tumor-bearing mouse models. Thus, similar to other HER2-targeted products, DS-8201a is expected to demonstrate efficacy in treating HER2-expressing tumors.

In nonclinical toxicology studies, the intestinal toxicity, hematopoietic system toxicity, pulmonary toxicity, testicular toxicity, skin toxicity, and renal toxicity were found in association with the administration of DS-8201a. In addition to these toxicities, similar to other products of the same class, the possibility of cardiotoxicity, hepatotoxicity, embryo-fetal toxicity, or phototoxicity occurring in subjects receiving DS-8201a cannot be excluded. As with any therapeutic antibodies, there is a possibility of infusion related reactions, immune responses causing allergic or anaphylactic reactions of DS-8201a.

Based on the efficacy and safety data observed in the nonclinical studies and the information from other products of the same class, the benefit-risk balance supports clinical development of DS-8201a.

1.4. Population, Route, Dosage, Dosage Regimen, Treatment Period

DS-8201a will be administered as an IV solution. Subjects with solid malignant tumors will receive DS-8201a on Day 1 of Q3W. An alternative drug administration schedule for dose escalation may be considered after establishing the MTD and/or RP2D of DS-8201a using the above schedule, and upon review of the human safety, and PK parameters by the Investigators and Sponsor (Refer to Section 3 and 4 for details.).

Subjects enrolled in Part 1, Part 2a, Part 2b, Part 2c and Part 2d will receive FL-DP1 drug product, and those enrolled in Part 2e will receive FL-DP2 drug product.

1.5. Compliance Statement, Ethics and Regulatory Compliance

This study will be conducted in compliance with the protocol, the ethical principles that have their origin in the Declaration of Helsinki, the International Conference on Harmonisation (ICH) consolidated Guideline E6 for Good Clinical Practice (GCP) (CPMP/ICH/135/95), and applicable regulatory requirement(s) including the following:

- US Food and Drug Administration (FDA) GCP Regulations: Code of Federal Regulations (CFR) Title 21, parts 11, 50, 54, 56 and 312, as appropriate.
- Japanese Ministry of Health, Labor and Welfare Ordinance No. 28 of 27 March, 1997.
- The Act on Securing Quality, Efficacy and Safety of Pharmaceuticals, Medical Devices, Regenerative and Cellular Therapy Products, Gene Therapy Products, and Cosmetics.
- Other applicable local regulations.

1.5.1. Subject Confidentiality

The Investigators and the Sponsor will preserve the confidentiality of all subjects taking part in the study, in accordance with GCP and local regulations.

The Investigator must ensure that the subject's anonymity is maintained. On the Electronic Case Report Forms (eCRF) or other documents submitted to Sponsor and/or its contract research organization (CRO), subjects should be identified by a unique Subject Number as designated by the Sponsor. Documents that are not for submission to Sponsor or CRO (eg, signed ICF) should be kept in strict confidence by the Investigator.

In compliance with ICH GCP Guidelines, it is required that the Investigator and institution permit authorized representatives of the company, of the regulatory agency(s), and the Independent Ethics Committee (IEC) or Institutional Review Board (IRB) direct access to review the subject's original medical records for verification of study-related procedures and data. The Investigator is obligated to inform the subject that his/her study-related records will be reviewed by the above named representatives without violating the confidentiality of the subject.

1.5.2. Informed Consent Procedure

Before a subject's participation in the study, it is the Investigator's responsibility to obtain freely given consent, in writing, from the subject after adequate explanation of the aims, methods, anticipated benefits, and potential hazards of the study and before any protocol-specific screening procedures or any study drugs are administered. The written ICF should be prepared in the local language(s) of the potential subject population.

In obtaining and documenting informed consent, the Investigator should comply with the applicable regulatory requirements, and should adhere to GCP and to the ethical principles that have their origin in the Declaration of Helsinki. The consent form and any revision(s) should be approved by the IRB prior to being provided to potential subjects.

The subject's written informed consent should be obtained prior to his/her participation in the study, and should be documented in the subject's medical records. The ICF should be signed and personally dated by the subject, and by the person who conducted the informed consent discussion (not necessarily the Investigator). The original signed ICF should be retained in accordance with institutional policy, and a copy of the signed consent form should be provided to the subject. The date that informed consent was given should be recorded on the eCRF.

If the subject cannot read, then according to ICH GCP Guideline, Section 4.8.9, an impartial witness should be present during the entire informed consent discussion. This witness should sign the ICF after the subject has orally consented to the subject's participation and, if possible, signed the ICF. By signing the ICF, the witness attests that the information in the ICF and any other written information was adequately explained to and apparently understood by the subject and that informed consent was freely given by the subject.

For studies in the United States, an additional consent is required for the HIPAA.

Suggested model text for the ICF for the study and any applicable subparts (genomic, PK, etc) is provided in the Sponsor ICF template for the Investigator to prepare the documents to be used at his or her site.

The consent form should be signed and personally dated by the subject or legally acceptable representative prior to his/her participation in the study.

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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Dose-Limiting Toxicities

3.1.7. Dose-limiting Toxicities

A DLT is defined as any TEAE not attributable to disease or disease-related processes that occurs during the DLT evaluation period (Day 1 to Day 21 in Cycle 1 of Part 1) and is grade 3 or above according to NCI-CTCAE version 4.0, with the exceptions as defined below:

For hematological toxicities, a DLT is defined as follows:

- Grade 4 neutrophil count decreased lasting > 7 days, neutrophil count decreased that requires standard therapies, such as colony stimulating factor (CSF)
- Grade ≥ 3 febrile neutropenia
- Grade 4 anemia
- Grade 4 platelet count decreased
- Grade ≥ 3 platelet count decreased lasting > 7 days
- Grade ≥ 3 platelet count decreased with clinically significant hemorrhage
- Grade 4 lymphocyte count decreased lasting ≥ 14 days

For hepatic organ toxicities, a DLT is defined as follows:

- Grade 4 AST or ALT increased
- AST or ALT > 5 × ULN (Grade 3), if accompanied by Grade ≥ 2 blood bilirubin increased
- In subjects without liver metastases, AST or ALT > 5 × ULN (Grade 3) lasting > 3 days
- In subjects with liver metastases, AST or ALT > 5 × ULN (Grade 3) lasting > 3 days, if the baseline level was ≤ 3 × ULN (Grade ≤ 1)
- In subjects with liver metastases, AST or ALT > 8 × ULN lasting > 3 days, if the baseline level was > 3 × ULN (Grade ≥ 2)

For non-hematological, non-hepatic major organ toxicities, a DLT is defined as follows:

- Symptomatic CHF
- LVEF decline to < 40% or > 20% drop from baseline
- Other grade ≥ 3 non-hematological, non-hepatic major organ toxicities

The following TEAEs are NOT considered DLTs:

- Grade 3 fatigue lasting < 7 days
- Grade 3 nausea, vomiting, diarrhea, or anorexia that has resolved to grade ≤ 2 within 3 days
- Isolated laboratory findings not associated with signs or symptoms including Grade 3/4 ALP increased, hyperuricemia, serum amylase increased, and lipase increased, and Grade 3 hyponatremia lasting < 72 hours developed from grade 1 at baseline
- Grade 3 lymphocyte count decreased

If any of the above toxicities is observed during the DLT evaluation period, whether the toxicity is regarded as DLT will be determined based on consultation between the Investigator and Sponsor.

In addition, with regard to other toxicities that hinder the conduct of the scheduled study treatment or anemia with blood transfusion, whether they are regarded as DLT will be determined based on consultation between the Investigator and Sponsor.
The premedication, which is a treatment before study drug administration to avoid AE, is prohibited during the DLT evaluation period.

However, the supportive therapy for the treatment is permitted after study drug administration:

For example:

- Neutrophil count decreased: CSF administration
- Nausea, vomiting: Antiemetics

3.1.8. Maximum Tolerated Dose and Recommended Phase 2 Dose Definition

Once the dose escalation stopping criteria are met, the MTD estimated by mCRM + EWOC will be the dose with the highest posterior probability of the DLT rate in the target DLT rate interval of (16%, 33%] among all doses fulfilling the overdose control constraint: there is less than 25% probability of a DLT rate > 33% (probability for excessive or unacceptable toxicity) (Section 3.2.1.2.1). The final MTD for each dosing schedule will be decided based on considerations of the respective MTDs estimated by the mCRM and on an overall assessment of safety data from subsequent cycles and PK information collected at all different doses tested. Upon determining the final MTD of the original Q3W and/or alternative dosing schedules, dosing regimen(s) will be selected for further evaluation in Part 2 (Dose Expansion). This regimen given at the final MTD determined in Part 1 is referred to as the "RP2D." RP2D can be several dose levels depending on the risk-benefit of study drug.

3.1.9. Management of Subjects With Adverse Events of Interests

3.1.9.1. Infusion Related Reactions

If an infusion related reaction is observed during DS-8201a administration, the infusion speed should be reduced or the infusion should be interrupted or discontinued based on the symptoms. Action taken for the infusion related reaction is referred according to the following Table 3.1.

Please see Section 9.3.3 for clinical summary and further management guidance.

Table 3.1: Actions Taken for Infusion Related Reaction

NCI-CTCAE Grade	Actions
Grade 1 Mild transient reaction: : infusion interruption not indicated : intervention not indicated	• If infusion related reaction is observed during administration, the infusion speed should be reduced by 50% and subjects should be closely monitored. • If no other reactions appear, the subsequent infusion speed could be resumed at the initial planned speed.

Management of AESI → Infusion-related reactions

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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AESI → Cardiac toxicity

Table 3.1: Actions Taken for Infusion Related Reaction (Continued)

NCI-CTCAE Grade	Actions
Grade 2 Therapy or infusion interruption indicated but responds promptly to symptomatic treatment (eg, antihistamines, nonsteroidal anti-inflammatory drugs (NSAIDs), narcotics, IV fluids); prophylactic medications indicated for ≤ 24 hours	• Administration of DS-8201a should be interrupted briefly. • Symptomatic treatment (eg, antihistamines, NSAIDs, narcotics, IV fluids) • If the event resolves or improves to grade 1, infusion can be re-started at a 50% reduced infusion speed. • Subsequent administrations should be conducted at the reduced speed.
Grade 3 Prolonged (eg, not rapidly responsive to symptomatic medication and/or brief interruption of infusion); recurrence of symptoms following initial improvement; hospitalization indicated for clinical sequelae	• Administration of DS-8201a should be discontinued immediately and permanently. • Urgent intervention indicated. Antihistamines, steroids, epinephrine, bronchodilators, vasopressors, intravenous fluid therapy, oxygen inhalation etc., should be administered.
Grade 4 Life-threatening consequences; urgent intervention indicated	

3.1.9.2. Cardiac Toxicity

Taking into consideration of the results of the nonclinical toxicology studies of DS-8201a and MAAA-1181 and the nonclinical and clinical information that are available for trastuzumab and T-DM1, the Investigator will check the LVEF to monitor for cardiac toxicity.

LVEF will be monitored by either ECHO or MUGA at the screening, BI of cycle 2, 3, and every 2 cycles thereafter until the end of treatment (EOT) (eg, Day 1 in Cycle 3, 5, 7, 9, 11...). Subjects with confirmed symptomatic cardiac dysfunction will be discontinued from the study. A symptomatic decline in LVEF will be handled with the algorithm shown in Table 3.2.

Please see Section 9.3.1 for clinical summary and further management guidance.

Table 3.2: Actions Taken for Cardiac Toxicity

Symptomatic CHF	LVEF < 40% or > 20% drop from baseline	LVEF 40% to ≤ 45% and decrease is ≥ 10% from baseline	LVEF 40% to ≤ 45% and decrease is < 10% from baseline	LVEF > 45%
Discontinue DS-8201a	Interrupt DS-8201a dosing Repeat LVEF assessment within 3 weeks. If LVEF < 40% or > 20% drop from	Interrupt DS-8201a dosing Repeat LVEF assessment within 3 weeks. If LVEF has not recovered to within	Continue treatment with DS-8201a Repeat LVEF assessment within 3 weeks.	Continue treatment with DS-8201a

Symptomatic CHF	LVEF < 40% or > 20% drop from baseline	LVEF 40% to ≤ 45% and decrease is ≥ 10% from baseline	LVEF 40% to ≤ 45% and decrease is < 10% from baseline	LVEF > 45%
	baseline is confirmed, discontinue DS-8201a.	10% from baseline, discontinue DS-8201a.		

CHF = Congestive Heart Failure; LVEF = Left Ventricular Ejection Fraction

3.1.9.3. Pulmonary Toxicity

ILD/pneumonitis is considered an important identified risk based on a comprehensive cumulative review of the available safety data from the DS8201-A-J101 clinical study as well as the results of potential ILD/pneumonitis cases reviewed by the independent ILD Adjudication Committee (AC), available data from recent epidemiology/literature, biological plausibility, and safety information from drugs of similar class. Refer to the current IB for a summary of preliminary clinical study data.

If a subject develops an acute onset of new or worsening pulmonary or other related signs/symptoms such as dyspnea, cough or fever, rule out interstitial lung disease (ILD) /pneumonitis.

If the AE is confirmed to have an etiology other than ILD/pneumonitis, follow the management guidance outlined in the "Other Non-Laboratory Adverse Events" dose modification section below.

If the AE is suspected to be ILD/pneumonitis, treatment with study drug should be interrupted pending further evaluations.

Evaluations should include:

- high resolution CT
- pulmonologist consultation
- pulmonary function tests and pulse oximetry (SpO_2)
- arterial blood gases if clinically indicated
- one blood sample collection for PK and exploratory biomarker analysis as soon as ILD/pneumonitis is suspected, if feasible.

Other tests could be considered, as needed.

As soon as ILD/pneumonitis is suspected, corticosteroid treatment should be started promptly as per clinical treatment guidelines (Kubo K, et al 2013 for guidance). [18]

If the AE is confirmed to be ILD/pneumonitis, follow the management guidance as outlined in Table 3.3.

All events of ILD regardless of severity or seriousness will be followed until resolution including after drug discontinuation.

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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Symptomatic CHF	LVEF < 40% or > 20% drop from baseline	LVEF 40% to ≤ 45% and decrease is ≥ 10% from baseline	LVEF 40% to ≤ 45% and decrease is < 10% from baseline	LVEF > 45%
	baseline is confirmed, discontinue DS-8201a.	10% from baseline, discontinue DS-8201a.		

CHF = Congestive Heart Failure; LVEF = Left Ventricular Ejection Fraction

3.1.9.3. Pulmonary Toxicity

ILD/pneumonitis is considered an important identified risk based on a comprehensive cumulative review of the available safety data from the DS8201-A-J101 clinical study as well as the results of potential ILD/pneumonitis cases reviewed by the independent ILD Adjudication Committee (AC), available data from recent epidemiology/literature, biological plausibility, and safety information from drugs of similar class. Refer to the current IB for a summary of preliminary clinical study data.

If a subject develops an acute onset of new or worsening pulmonary or other related signs/symptoms such as dyspnea, cough or fever, rule out interstitial lung disease (ILD) /pneumonitis.

If the AE is confirmed to have an etiology other than ILD/pneumonitis, follow the management guidance outlined in the "Other Non-Laboratory Adverse Events" dose modification section below.

If the AE is suspected to be ILD/pneumonitis, treatment with study drug should be interrupted pending further evaluations.

Evaluations should include:

- high resolution CT
- pulmonologist consultation
- pulmonary function tests and pulse oximetry (SpO_2)
- arterial blood gases if clinically indicated
- one blood sample collection for PK and exploratory biomarker analysis as soon as ILD/pneumonitis is suspected, if feasible.

Other tests could be considered, as needed.

As soon as ILD/pneumonitis is suspected, corticosteroid treatment should be started promptly as per clinical treatment guidelines (Kubo K, et al 2013 for guidance). [18]

If the AE is confirmed to be ILD/pneumonitis, follow the management guidance as outlined in Table 3.3.

All events of ILD regardless of severity or seriousness will be followed until resolution including after drug discontinuation.

Please see Section 9.3.2 for clinical summary and further management guidance.

Table 3.3: Actions Taken for Study Drug Induced ILD or pneumonitis

NCI-CTCAE Grade	Actions
Grade 1	The administration of DS-8201a must be interrupted for any ILD events regardless of grade. For Grade 1 events, DS-8201a can be restarted only if the event is fully resolved to Grade 0: • If resolved in ≤ 28 days from date of onset, maintain dose • If resolved in > 28 days from date of onset, reduce dose 1 level
Grade 2, 3 or 4	Permanently discontinue subject from study treatment.

AESI → Pulmonary toxicity

3.2. Selection of Doses

3.2.1. Experimental Treatments

The study will enroll subjects into cohorts with dose escalation by mCRM with EWOC principle as outlined in Section 3.2.1.2. The starting dose will be 0.8 mg/kg (See Section 1.1.1.4.5 for justification of the human starting dose).

Upon completion of Cycle 1 of Part 1 (Dose Escalation) and determination of the MTD/RP2D, Part 2 (Dose Expansion) will begin. In Part 1, the second subject in each dosing level should start dosing at least 24 hours after the initial dosing of the first subject to check the acute toxicity such as infusion-related reaction. Subjects in Part 2 will receive DS-8201a at the MTD/RP2D defined in Section 3.2.1.2.

Subjects in Part 1, Part 2a, Part 2b, Part 2c and Part 2d will receive FL-DP1 drug product, and those in Part 2e will receive FL-DP2 drug product.

3.2.1.1. For the Study

See Sections 1.1.1.4.5 and 1.2.

3.2.1.2. For Individual Subjects

3.2.1.2.1. Part 1 (Dose Escalation)

Dose escalation of DS-8201a to determine the MTD/RP2D will be guided by the mCRM using a BLRM following EWOC principle. Dose escalation will begin in subjects with advanced breast cancer and gastric or gastroesophageal junction adenocarcinoma. At least 18 subjects will be enrolled in Part 1. More than 16% of subjects (eg, 1 per 6 subjects) are expected to be HER2 expression (IHC 2+ or 3+). The proposed human starting dose of DS-8201a is 0.8 mg/kg, based on 1/12 HED of HNSTD in monkey (30 mg/kg).

During Part 1, an initial dose of DS-8201a will be infused intravenously into each subject for approximately 90 minutes on Day 1 of Cycle 1. A 21-day observation period (Cycle 1) will then occur, at the end of which all relevant safety data will be reviewed. Upon completion of Cycle 1, subjects may continue to receive DS-8201a Q3W(1 cycle) at the discretion of the Investigators, until unacceptable toxicity, PD, or withdrawal of consent. If there is no infusion related reaction

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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Drug Substance Drug Product

Labelling Packaging Preparation (In-use stability)

5. TREATMENTS ADMINISTERED

5.1. Investigational Products

The head of the study center in Japan and Investigator in the United States must ensure that the investigational product will be used only in accordance with the protocol.

Two types of DS-8201a drug product, FL-DP1 and FL-DP2, will be supplied in this study. FL-DP1 drug product will be used for Part 1, Part 2a, 2b, 2c and 2d. FL-DP2 drug product will be used for Part 2e.

The FL-DP1 clinical material is provided for infusion as a [redacted] of DS-8201a in a glass vial. FL-DP2 clinical material is provided for infusion as a [redacted] of DS-8201a in a glass vial.

It is supplied in single-use glass vials of sterile drug product solution.

5.1.1. Method of Assigning Subjects to Treatments and Blinding

5.1.1.1. Randomization

Not applicable.

5.1.1.2. Blinding

Both parts of the study are open-label and no blinding will be performed.

5.1.2. Method of Assessing Treatment Compliance

All drugs used for the study treatment will be administered by the Investigator or other designated study personnel. Therefore, treatment compliance will be guaranteed as long as the subject attends each visit for the administration of study treatment.

5.1.3. Labeling and Packaging

DS-8201a will be supplied by Sponsor. Study drug supplies will be packaged in cartons containing [redacted] FL-DP1 [redacted] and [redacted] FL-DP2 [redacted]. The packaging will be clearly labeled "For Clinical Study Use Only," and will show the display name of the investigational product, the investigational product manufacturing code, storage condition and other required information in accordance with local regulations.

5.1.4. Preparation

The drug for IV infusion is prepared by dilution of the required volume of the drug product calculated based on the subject's body weight in a volume of 100 mL. Prepared medicinal solutions should be used immediately. The preparation will be conducted in accordance with the pharmacy instructions provided by the Sponsor. Procedures for proper handling and disposal of anticancer drugs should be followed in compliance with the standard operating procedures (SOPs) of the study site.

5.1.5. Storage

Drug supplies must be stored in a secure, limited access storage area under the storage conditions listed below:

- Stored below -15°C (protected from light).

If storage conditions are not maintained per specified requirements, Sponsor or CRO should be contacted.

Storage

5.1.6. Drug Accountability

When a drug shipment is received, the Investigator or designee will check the amount and condition of the drug, check for appropriate local language in the label, drug expiration date, and sign the Receipt of Shipment Form provided. The Receipt of Shipment Form should be signed and the original Form will be retained at the site. In addition, the Investigator or designee shall contact Sponsor as soon as possible if there is a problem with the shipment.

A Drug Accountability Record will be provided for the investigational product. The record must be kept current and should contain, the dates and quantities of drug received, subject's (identification number and/or initials or supply number as applicable), for whom the investigational product was dispensed, the date and quantity of investigational product dispensed and remaining, if from individual subject drug units as well as the initials/seal of the dispenser.

At the end of the study, or as directed, all unused DS-8201a will be returned to a designee as instructed by Sponsor. Investigational Product will be returned only after the study monitor has completed a final inventory to verify the quantity to be returned. The return of Investigational Product must be documented and the documentation included in the shipment. At the end of the study, a final Investigational Product reconciliation statement must be completed by the Investigator or designee and provided to the Sponsor. Unused drug supplies may be destroyed by the supervisor of investigational products in Japan and Investigator in the United States when approved in writing by Sponsor and Sponsor has received copies of the site's drug handling and disposition SOPs.

All investigational product inventory forms must be made available for inspection by a Sponsor authorized representative or designee and regulatory agency inspectors. The supervisor of investigational products in Japan and Investigator in the United States is responsible for the accountability of all used and unused study supplies at the site.

5.1.7. Retention Samples

Not applicable.

5.2. Concomitant Medications

Medications used from Day 1 of Cycle 1 to 28 days (– 7 days) after the last administration of DS-8201a will be recorded. All concomitant medications will be recorded on the eCRF.

5.2.1. Prohibited Concomitant Medications/Activities

The following medications and products will be prohibited:

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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PK Sampling Timepoints

Table 6.1: Pharmacokinetic Sampling Time Points (Part 1 and Part 2)

Cycle	Day	Sampling Time Point (Acceptable Range)
Cycle 1	Day 1	BI (– 8 hours) EOI: Within 15 minutes after EOI 2 hours after the start of administration (± 15 minutes) 4 hours after the start of administration (± 15 minutes) 7 hours after the start of administration (± 15 minutes)
	Day 2	24 hours after the start of administration (± 2 hours)
	Day 4	72 hours after the start of administration (± 2 hours for Par 1, ± 1 day for Part 2)
	Day 8	7 days after the start of administration (± 1 day)
	Day 15	14 days after the start of administration (± 1 day)
	Day 22	If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample 21 days after the start of administration (± 2 days)
Cycle 2	Day 1	BI (– 8 hours) If blood sample is collected on Day 22 of Cycle 1, the blood sample will be collected at BI on Day 1 of Cycle 2 if possible. EOI: Within 15 minutes after EOI
	Day 22	If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample on 21 days after the start of administration (± 2 days)
Cycle 3	Day 1	BI (– 8 hours) If blood sample is collected on Day 22 of Cycle 2, the blood sample will be collected at BI on Day 1 of Cycle 3 if possible. EOI: **Part 1 (Dose escalation)** Within 15 minutes after EOI 4 hours after the start of administration (± 15 minutes) **Part 2 (Dose expansion)** Within 15 minutes after EOI
	Day 8	7 days after the start of administration (± 2 days)
	Day 15	14 days after the start of administration (± 2 days)
	Day 22	If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample on 21 days after the start of administration (± 2 days)

Table 6.1: Pharmacokinetic Sampling Time Points (Part 1 and Part 2) (Continued)

Cycle	Day	Sampling Time Point (Acceptable Range)
Cycle 4, 6, 8	Day 1	BI (– 8 hours) If blood sample is collected on Day 22 of Cycle 3, the blood sample will be collected at BI on Day 1 of Cycle 4 if possible. EOI: Within 15 minutes after EOI

6.8.4. Pharmacodynamic (PDy)

Not applicable.

6.8.5. Exploratory Biomarker

Based on the evaluation of pre-treatment tumor sample, the levels of HER2 will be assessed by IHC and FISH. The exploratory biomarkers (eg, HER2ECD, HER2 by HERmark, cTnI testing by central lab) will be measured.

Blood samples of approximately 6 mL for HER2ECDs analyses will be collected at the time points specified in Table 6.2. This is applicable until protocol version 11.0. No further blood sampling for HER2ECD is necessary once protocol version 12.0 is applied for a subject.

Table 6.2: HER2ECD Sampling Time Points

Cycle	Day	Sampling Time Point (Acceptable Range)
Screening	-	Within 7 days before registration
After Cycle 3 at every 2 cycles (eg, Cycle 3, 5, 7, 9, 11...)	Day 1	BI (– 8 hours)
EOT	-	The date Investigator decides the discontinuation of the study treatment (+ 7 days).

In addition to troponin testing by study site, blood samples of approximately 6 mL for cTnIs analyses by central lab will be collected at the time points specified in Table 6.3.

Table 6.3: cTnI Sampling Time Points

Cycle	Day	Sampling Time Point (Acceptable Range)
Screening	-	Within 7 days before registration
Every Cycle	Day 1	EOI
EOT	-	The date Investigator decides the discontinuation of the study treatment (+ 7 days).
F/U	-	28 days (– 7 days) after the last administration of DS-8201a.

Blood samples of approximately 20 mL for immune monitoring analysis will be collected from Japanese subjects in Part 2d at the time points specified in Table 6.4.

PD Sampling Timepoints

Efficacy: HER2
Safety: cTnI

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

Immune Monitoring Sampling Timepoints

Table 6.4: Immune Monitoring Sampling Time Points (Japanese subjects in Part 2d only)

Cycle	Day	Sampling Time Point (Acceptable Range)
Screening – BI at Cycle 1 Day 1	-	-
Cycle 1	Day 2	-
	Day 4	-
	Day 8	-
	Day 22	If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample 21 days after the start of administration (± 2 days)
Cycle 2	Day 1	BI. This is not necessary if blood samples are obtained on Cycle 1, Day 22.
	Day 22	If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample 21 days after the start of administration (± 2 days)
Cycle 3	Day 1	BI. This is not necessary if blood samples are obtained on Cycle 2, Day 22.
Cycle 4 and subsequent cycles	Day 1	BI. One time during treatment period.
EOT	-	The date Investigator decides the discontinuation of the study treatment (+ 7 days).

cfDNA Sampling Timepoints

Blood samples of approximately 20 mL for cfDNA analysis will be collected at the time points specified in Table 6.5.

Table 6.5: Cell Free DNA Sampling Time Points (Optional)

Cycle	Day	Sampling Time Point (Acceptable Range)
Screening – BI at Cycle 1 Day 1	-	-
EOT	-	The date Investigator decides the discontinuation of the study treatment (+ 7 days).

Instructions for the handling and shipping of samples are included in a separate document (eg, laboratory manual).

6.8.6. Anti-drug Antibody

Blood samples for ADA of approximately 4 mL analyses will be collected at the time points specified in Table 6.6. Serum concentrations of DS-8201a and/or total anti-HER2 antibody may be measured using the same ADA samples for purpose of anti-drug antibody assessment.

Instructions for the handling and shipping of serum samples are included in a separate document (eg, laboratory manual).

Table 6.6: ADA Sampling Time Points

Cycle	Day	Sampling Time Point (Acceptable Range)
Cycle 1	Day 1	BI (– 8 hours)
	Day 8	± 1 day
Cycle 2	Day 1	BI (– 8 hours)
After Cycle 4 at every 2 cycles (eg, Cycle 4, 6, 8, 10, 12...)	Day 1	BI (– 8 hours)
EOT	-	The date Investigator decides the discontinuation of the study treatment (+ 7 days).
F/U*	-	28 days (– 7 days) after the last study drug administration or until starting new anticancer treatment, whichever comes first.

ADA Sampling Timepoints

* For subjects with positive ADA at F/U visit, additional serum ADA samples may be collected every 3 months (± 1 month) up to 1 year from the last dose of study drug, or if the ADA becomes negative, or if ADA titer becomes less than baseline (applicable when pre-existing ADA is observed), or if the subject starts another therapy for cancer, or withdraws consent from the study, whichever occurs first.

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

ENHERTU®: First-In-Human Study (NCT02564900)

17.6. Schedule of Events (Part 1 and Part 2)

	SCR	Cycle 1							Cycle 2					Cycle 3					Cycle 4 and subsequent cycles		EOT [a]	F/U [b]
		Day 1		Day 2	Day 4	Day 8	Day 15	(Day 22)	Day 1		Day 8	Day 15	(Day 22)	Day 1		Day 8	Day 15	(Day 22)	Day 1			
		BI	EOI			(± 1 day)	(± 1 day)	(± 2 days)	BI	EOI	(± 2 days)	(± 2 days)	(± 2 days)	BI	EOI	(± 2 days)	(± 2 days)	(± 2 days)	BI	EOI		
Informed consent	X																					
Administration DS-8201a		X							X[n]					X[n]					X[n]			
Demographic information	X																					
Vital sign	X[c]	X[d]	X	X	X	X	X		X[d]	X	X	X		X[d]	X	X	X		X[d]		X	X
Physical examination	X[c]	X[d]							X[d]					X[d]					X[d]		X	X
SpO_2		X[d]							X[d]					X[d]					X[d]		X	
I/E Criteria	X																					
Height	X																					
Weight, ECOG PS	X[c]	X[d]							X[d]					X[d]					X[d]		X	X
Laboratory tests	X[c]	X[d]		X		X	X		X[d]		X	X		X[d]		X	X		X[d]		X	X
PK		X[e]	X[f,g]	X[h]	X[h]	X	X	(X)[i]	X[e]	X[f]			(X)[i]	X[e]	X[f,j]	X	X	(X)[i]	X[e]	X[f]		
ADA		X[k]				X			X[k]										X[k]		X	X[q]
HIV antibody test	X[l]																					
HER2ECD (blood)	X[c]													X[m]							X	
Troponin (blood)	X[c]		X[n]							X[n]					X[n]					X[n]	X	X
ECHO or MUGA (LVEF)	X[o]								X[g]					X[g,m]							X	
12-lead ECG in triplicate	X[c]	X[d]	X[p]	X		X	X		X[d]	X				X[d]					X[d]		X	
Pregnancy test	X[c]																				X	

	SCR	Cycle 1							Cycle 2					Cycle 3					Cycle 4 and subsequent cycles		EOT [a]	F/U [b]
		Day 1		Day 2	Day 4	Day 8	Day 15	(Day 22)	Day 1		Day 8	Day 15	(Day 22)	Day 1		Day 8	Day 15	(Day 22)	Day 1			
		BI	EOI			(± 1 day)	(± 1 day)	(± 2 days)	BI	EOI	(± 2 days)	(± 2 days)	(± 2 days)	BI	EOI	(± 2 days)	(± 2 days)	(± 2 days)	BI	EOI		
Tumor assessment	X[c]	Every 6 weeks (± 7 days) in the first 24 weeks after Day 1 of Cycle 1, and thereafter every 12 weeks (± 7 days)																			X	
Ophthalmologic assessments	X[c]								X[d] At Day 1 in Cycle 2 and every 4 cycles from Cycle 5 to the EOT (eg, Day 1 in Cycle 2, 5, 9, 13...)												X	
IHC/FISH, HERmark (tumor sample)	X																					
cfDNA (blood)	(X)																				(X)	
For Part 2d only: Immune monitoring (blood)	X			X	X	X		(X)[s]	X				(X)[s]	X					X[s]		X	
For Part 2d only: Exploratory biomarker (tumor sample)	X	(X)[t] One time during Day 8 to 22 (prior to the subsequent study drug administration) of any cycle																			(X)[t]	
Concomitant medications	X																					
AEs	X[c]																					

SCR: screening, F/U: follow-up, BI: before infusion, EOI: end of infusion, EOT: end of treatment

[a] The date Investigator decides the discontinuation of the study treatment (+ 7 days)
[b] 28 days (− 7 days) after the last study drug administration or before starting new anticancer treatment, whichever comes first.
[c] Latest data within 7 days before registration. Troponin (preferably high-sensitivity troponin T) testing by study site and cTnI testing by central lab.
[d] Latest data within 3 days before administration
[e] Within 8 hours BI on Day 1 of each cycle until Cycle 4 and then every 2 cycles until Cycle 8 (eg, Cycle 1, 2, 3, 4, 6, 8).
[f] Within 15 minutes at EOI on Day 1 of each cycle until Cycle 4 and then every 2 cycles until Cycle 8 (eg, Cycle 1, 2, 3, 4, 6, 8).
[g] 2, 4, 7 hours after the start of administration (± 15 minutes)
[h] 24 (± 2 hours) and 72 hours (± 2 hours for Part 1, ± 1 day for Part 2) after the start of administration
[i] If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample for PK analysis
[j] Part 1 (Dose escalation): Within 15 minutes at EOI, 4 hours after the start of administration (± 15 minutes), Part 2 (Dose expansion): Within 15 minutes at EOI

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

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k Within 8 hours BI on Day 1 in Cycle 1 and 2, and then every 2 cycles until the end (eg, Cycle 1, 2, 4, 6, 8, 10, 12...)

l Latest data within 90 days before registration. HIV antibody test must be performed for Japanese subjects, and is optional for US subjects unless required by local regulations or IRB.

m Before administration for ECHO/MUGA and within 8 hours BI for HER2ECD. At every 2 cycles from Cycle 3 (eg, Cycle 3, 5, 7, 9...). For HER2ECD, this is applicable until protocol version 11.0. No further blood sampling for HER2ECD is necessary once protocol version 12.0 is applied for a subject.

n EOI for cTnI testing by central lab and 2 to 3 hours after EOI for troponin (preferably troponin T) testing at study site. Depending on the result of troponin testing at study sites 2 to 3 hours after EOI, additional testing(s) is needed.
If troponin levels are consistent with myocardial infarction as defined according to manufacturer (CTCAE Grade 3), perform ECG testing in triplicate, repeat troponin testing 6 hours (±1 hr) and 12 hours (±1 hr) after initial troponin test was drawn, and follow institutional guidelines. If troponin levels are above the upper limit of normal and below the level of myocardial infarction as defined by the manufacturer (CTCAE Grade 1), repeat troponin testing at 3 hours (±1 hr) after initial troponin test was drawn. If troponin level at 3 hours (6 hours post-infusion): Significantly increases per institutional guidelines, then repeat troponin testing at 6 hours (±1 hr) and follow institutional guidelines. Otherwise, repeat troponin testing at 6 hours (±1 hr) or at 24 hours (±2 hours) after initial troponin test. If troponin levels are above the upper limit of normal at baseline and below the level of myocardial infarction as defined by the manufacturer (CTCAE Grade 1), no repeat testing is required after the first EOI 3-hour troponin test if the troponin level is not Grade 3.

o Latest data within 28 days before registration

p Within 30 minutes after end of infusion (EOI), and 2 to 4 hours after the start of administration

q For subjects with positive ADA at F/U visit, additional serum ADA samples may be collected every 3 months (± 1 month) up to 1 year from the last dose of study drug, or if the ADA becomes negative, or if ADA titer becomes less than baseline (applicable when pre-existing ADA is observed), or if the subject starts another therapy for cancer, or withdraws consent from the study, whichever occurs first.

r For suspected ILD/pneumonitis, treatment with study drug should be interrupted pending evaluation.
Evaluations should include:
- high resolution CT
- pulmonologist consultation
- pulmonary function tests and pulse oximetry (SpO_2)
- arterial blood gases if clinically indicated
- one blood sample collection for PK and exploratory biomarker analysis as soon as ILD/pneumonitis is suspected, if feasible.

Other tests could be considered, as needed.

s If the schedule on Day 1 of the next cycle is delayed for 3 days or more, including if the subject cannot continue onto the next cycle, collect blood sample for immune monitoring analysis.

t It is recommended to perform a tumor biopsy one time during Day 8 to 22 (prior to the subsequent study drug administration), and EOT if the subject discontinues from the study treatment due to PD.

u 21 days (+ 2 days) from latest administration date.

v One time one time during treatment period.

For suspected ILD/pneumonitis, study drug should be interrupted pending evaluation.

Interstitial Lung Disease Management

[Reference] https://clinicaltrials.gov/study/NCT02564900#more-information

의약품안전나라 > 임상시험정보 > 임상시험정보공개

- 엔허투
  - 2024-12-27 승인
  - 위암 3상



[Reference]
https://nedrug.mfds.go.kr/pbp/CCBBC01/nexacroPageOpen?approvalDtStart=2022-10-05&searchType=ST3&searchYn=true&searchKeyword=%EC%97%94%ED%97%88%ED%88%AC&localList=000&approvalDtEnd=2025-10-05&page=1&&clinicExamSeq=202400941&clinicExamNo=33375&receiptNo=20250110935&approvalDt=2024-12-27

엔허투: 임상시험승인현황 (DESTINY-Gastric05)

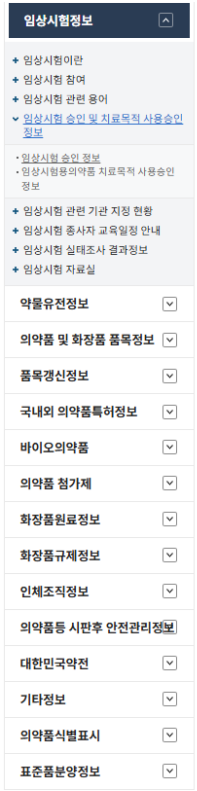
- 개요: 절제 불가능, 국소 진행성 또는 전이성 HER2 양성 위암 또는 위식도접합부 (GEJ)암이 있는 대상자에서 1차 치료로 트라스투주맙 데룩스테칸 (엔허투®) + 화학요법 +/- 펨브롤리주맙 대비 화학요법 + 트라스투주맙 +/- 펨브롤리주맙의 다기관, 무작위배정, 공개, 제3상 임상시험 (DESTINY-Gastric05)



[Reference]
https://nedrug.mfds.go.kr/pbp/CCBBC01/nexacroPageOpen?approvalDtStart=2022-10-05&searchType=ST3&searchYn=true&searchKeyword=%EC%97%94%ED%97%88%ED%88%AC&localList=000&approvalDtEnd=2025-10-05&page=1&&clinicExamSeq=202400941&clinicExamNo=33375&receiptNo=20250110935&approvalDt=2024-12-27

엔허투: 임상시험승인현황 (DESTINY-Gastric05)

- 임상시험용 의약품
- 연구설계
- 수행 및 평가방법
- 선정기준
- 제외기준





[Reference]
https://nedrug.mfds.go.kr/pbp/CCBBC01/nexacroPageOpen?approvalDtStart=2022-10-05&searchType=ST3&searchYn=true&searchKeyword=%EC%97%94%ED%97%88%ED%88%AC&localList=000&approvalDtEnd=2025-10-05&page=1&&clinicExamSeq=202400941&clinicExamNo=33375&receiptNo=20250110935&approvalDt=2024-12-27

엔허투: 임상시험승인현황 (DESTINY-Gastric05)

- 연구설계
  - 임상시험 예상기간
  - 목표 시험대상자 수
  - 최초 시험대상자 선정일
- 실시기관 등



개요 | 임상시험용 의약품 | 연구설계 | 수행 및 평가방법 | 선정기준 | 제외기준 | 실시기관 등

임상시험 단계	3상	임상시험 형태	Parallel
임상시험 예상기간	2025년 02월 ~ 2030년 02월	목표시험대상자수 (국내 대상자수)	726(93)
무작위 배정 여부	Randomized	눈가림 종류	Open
중재군 수	4	최초 시험대상자 선정일	2025-06-17
최종 시험대상자 선정일	- -	최종 시험대상자 관찰 종료일	- -
실제 시험 완료일	- -	결과보고 제출일	- -

개요 | 임상시험용 의약품 | 연구설계 | 수행 및 평가방법 | 선정기준 | 제외기준 | 실시기관 등

○ 임상시험 실시기관 및 시험책임자 14 건

순번	기관유형	실시기관명	시험책임자	과명, 직명
1	임상시험 실시기	분당서울대학교병원	이근욱	교수
2	임상시험 실시기	서울대학교병원	오도연	교수
3	임상시험 실시기	재단법인아산사회복지	류민희	교수
4	임상시험 실시기	칠곡경북대학교병원	김종광	교수
5	임상시험 실시기	충북대학교병원	한혜숙	교수
6	임상시험 실시기	국립암센터	김학균	교수
7	임상시험 실시기	화순전남대학교병원	배우균	교수
8	임상시험 실시기	전북대학교병원	송은기	교수

○ 임상시험 실시기관 및 시험책임자 상세

기관유형	● 임상시험실시기관 ○ 기타
실시기관명	분당서울대학교병원
기관소재지	경기도 성남시 분당구 구미로173번길 82 (구미동
시험책임자	이근욱
과명,직명	교수
실시기관 전화번호	-
모집공고 확인 홈페이지 주소	-

[Reference]
https://nedrug.mfds.go.kr/pbp/CCBBC01/nexacroPageOpen?approvalDtStart=2022-10-05&searchType=ST3&searchYn=true&searchKeyword=%EC%97%94%ED%97%88%ED%88%AC&localList=000&approvalDtEnd=2025-10-05&page=1&&clinicExamSeq=202400941&clinicExamNo=33375&receiptNo=20250110935&approvalDt=2024-12-27

요약: 임상시험의 기본원칙 (KGCP)

- 임상시험은 헬싱키선언에 근거한 윤리규정, 의약품 임상시험 관리기준 및 관계 법령에 따라 실시하여야 한다
- 임상시험에서 예측되는 위험과 불편을 충분히 고려하여 대상자 개인과 사회가 얻을 수 있는 이익이 그 위험과 불편보다 크거나 이를 정당화할 수 있다고 판단되는 경우에만 임상시험을 실시하여야 한다
- 과학과 사회의 이익보다 대상자의 권리・안전・복지를 먼저 고려하여야 한다
- 해당 임상시험용 의약품에 대한 임상 및 비임상 관련 정보는 실시하려는 임상시험에 적합한 것이어야 한다
- 임상시험은 과학적으로 타당하여야 하며, 임상시험 계획서는 명확하고 상세히 기술하여야 한다
- 임상시험은 식품의약품안전처장이 승인한 임상시험 계획서에 따라 실시하여야 한다

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do

요약: 임상시험의 기본원칙 (KGCP)

- 대상자에 대한 의학적 처치나 결정은 의사・치과의사 또는 한의사의 책임하에 이루어져야 한다
- **임상시험 수행에 참여하는 모든 사람들은 각자의 업무 수행에 필요한 적절한 교육과 훈련을 받고, 해당 업무 분야와 관련한 경험을 갖고 있어야 한다**
- 임상시험 참여 전에 모든 대상자로부터 자발적인 임상시험 참가 동의를 받아야 한다
- 모든 임상시험 관련 정보는 정확한 보고, 해석 및 확인이 가능하도록 기록・처리・보관되어야 한다
- 대상자의 신상에 관한 모든 기록은 비밀이 보장되도록 관계 법령에 따라 취급하여야 한다
- **임상시험용 의약품은 임상시험용의약품 제조 및 품질관리기준에 따라 관리되고, 식품의약품안전처장이 승인한 임상시험 계획서에 따라 사용되어야 한다**
- **임상시험은 품질보증이 이뤄질 수 있는 체계하에서 실시되어야 한다**

[Reference] https://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do


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